

Product Switching and Young Firm Dynamics*

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Abstract

We develop a model in which firms learn about their performance across product markets and adjust their product portfolios over the lifecycle. Firms collect and process information to extract signals, and their information-processing capacity improves over time. However, limited processing ability early in the lifecycle generates systematic biases in beliefs. These biases create distinctive product switching behavior of young firms, inducing wedges in within-firm resource allocation across products. As a result, learning frictions slow firms' reallocation toward high-performing products early in the lifecycle. This mechanism introduces a within-firm margin of adjustment that complements the standard cross-firm selection channel in shaping resource allocation and aggregate productivity. Using firm-product-level administrative data from the U.S. manufacturing sector, we show that firms are more likely to retain low-potential products and drop high-potential ones when they are young, consistent with the model's predictions. Improving firms' information-processing ability or reducing product search costs enhances allocative efficiency.

JEL Code: L11, L21, L25, L60, O31

Keywords: product add and drop, product switching, young firm dynamics, firm growth, resource allocation, aggregate productivity

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1 Introduction

Finding an optimal set of products to produce is important for how firms allocate resources. Firms add and drop products throughout their life-cycle, and the literature finds that product switching is prevalent and has a profound impact on a firm scope and performance (Bernard et al., 2010; Argente et al., 2018). When switching products, firms often remain close to their existing ones (e.g., Bernard et al., 2010; Boehm et al., 2022). However, firms also venture into entirely new industries, and such radical product switching can yield substantial performance gains and contribute to long-term success.¹ At the same time, many firms fail to introduce new products successfully, suggesting that they may excel in certain areas while struggling in others.²

These decisions may be particularly important when firms are young, as their production scope and specialty are not yet fully established. As a result, they may need to search more intensively to find the right products to produce. These dynamics have important aggregate implications, as the growth of young firms is a key driver of aggregate productivity and economic growth (Haltiwanger, 2012; Haltiwanger et al., 2013; Decker et al., 2014, 2016; Haltiwanger et al., 2016; Foster et al., 2018).

In this paper, we study firms’ product-switching decisions over the lifecycle and their implications for firm growth and resource allocation. We develop a model in which firms learn about how well their products perform in the market—how appealing products are to consumers—through production and sales, which we term product-specific expertise. Firms then use these beliefs to adjust their product portfolios over the lifecycle. Our framework highlights a novel mechanism: imperfect learning early in the lifecycle generates systematic biases in beliefs, which in turn shape production and product-switching decisions. We show that this mechanism leads to persistent differ-

¹For example, Samsung started as a grocery trading company, but by shifting its core business to electronics and semiconductor manufacturing, it eventually evolved into the globally recognized firm as is today. Similarly, 3M began as a mining venture but became renowned by diversifying into a wide range of products for worker safety, healthcare, and consumer products.

²For instance, Dyson and Apple both attempted to enter the electric car market but ultimately failed. Likewise, McDonald’s was unsuccessful in its attempt to sell pizza, despite potential complementarities in inputs and distribution networks.

ences in product choices across firms and over time, with implications for resource allocation across product markets. Using firm–product-level administrative data from the U.S. manufacturing sector, we provide evidence consistent with the model’s key predictions.

The model builds on the learning framework of [Jovanovic \(1982\)](#) and product switching behavior of multi-product firms in [Bernard et al. \(2010\)](#). Firms are born with heterogeneous types that determine productivity, product scope, and their baseline ability to process information and extract signals; this information-processing ability improves over time through experience and data accumulation. They are also endowed with time-invariant expertise in each product market, which is initially unobserved. Firms form prior beliefs about this expertise and update them over time using realized market-clearing prices and extracted signals through a Bayesian learning process. Based on their beliefs, firms decide whether to search for a new product to add or to switch existing ones, incurring a search cost.

The model generates rich lifecycle patterns in product-switching dynamics: (i) Biased and noisy beliefs early in the lifecycle. When firms are young, lower information-processing ability leads to noisier signals and greater reliance on the prior, resulting in systematic underestimation of high-potential products (high product-specific expertise) and overestimation of low-potential products, along with more diffuse posterior beliefs. (ii) Belief-driven patterns in production. These biased beliefs translate into a wedge between perceived and true product potential, leading firms to produce relatively less of high-potential products and relatively more of low-potential products. (iii) Systematic patterns in product-switching behavior. As a consequence, when firms are young, they are less likely to retain high-potential products and more likely to retain low-potential products, resulting in less precise product-portfolio decisions early in the lifecycle.

These mechanisms have novel aggregate implications for resource allocation. Product-portfolio adjustments by firms constitute an additional margin through which resources are allocated within and across product markets. Because firms’ beliefs are systematically biased and less precise early in the lifecycle, resources may be disproportion-

ately directed toward low-potential products and away from high-potential ones. Since learning and belief updating are gradual, and firms continuously enter and exit, these wedges persist and affect aggregate allocation, with implications for aggregate productivity.

The model provides a framework to quantify the contribution of this channel to aggregate outcomes and to evaluate policies that improve information-processing ability and decision-making. In particular, policies that enhance firms’ access to data, improve managerial practices, or reduce search costs can accelerate learning and product switching, thereby mitigating these wedges. More broadly, the framework can be used to study how interventions—such as support for struggling firms—affect product-portfolio adjustments and aggregate productivity.

Next, we use a detailed firm-product level dataset for the manufacturing sector from the U.S Census Bureau from 1982 to 2007 to test the model’s predictions. In this dataset, we define a “product” as a five-digit Standard Industrial Classification (SIC) category for the pre-2002 years and a seven-digit North American Industry Classification System (NAICS) category from 2002 onward. We construct product-firm pairs in each Census year. A key contribution of our empirical approach is the estimation of firms’ time-invariant product-specific expertise using firm–product fixed effects from real shipment values.³ To our knowledge, this represents a novel contribution of the paper.

When estimating time-invariant product-specific expertise, we control for time-varying compounding factors such as product life-cycle dynamics (e.g., [Argente et al., 2024](#)), customer capital accumulation, and transitory product-level shocks (e.g., fluctuations in market conditions) that may obscure the firm-specific ability. The resulting measure isolates a firm’s underlying capacity to produce and sell a given product, capturing the extent of its expertise in that product market.

Using this expertise measure, we empirically investigate firms’ product-switching be-

³Our approach closely follows [Bonhomme et al. \(2019\)](#), which estimates worker-firm complementarities in earnings.

havior and the optimization of their product portfolios over the firm lifecycle—particularly when firms are young—and examine how these decisions affect subsequent firm growth and performance. We document a set of empirical patterns that support the model’s predictions: firms generally drop products with low product-specific expertise more frequently. However, when firms are young, they exhibit a distinct pattern. They drop products less frequently overall, but are more likely to drop high-expertise products and less likely to drop low-expertise products. These findings are consistent with biased and less precise beliefs early in the lifecycle.

In addition, we find that product switching plays an important role in firm growth and performance. In particular, product churning—the simultaneous adding and dropping of products—is positively associated with multiple measures of firm performance, including firm growth in value added, employment, total factor productivity, and labor productivity. Furthermore, decomposing firm-level growth into the growth of continuing products, reallocation across continuing products, and product switching, we find that product switching accounts for a substantial share of firm growth, particularly among young firms. Combined with the observed product-switching patterns of young firms, these results suggest that firms move up the product-expertise ladder more slowly early in the lifecycle, which can slow firm growth.

Our findings indicate that firms face greater challenges in identifying and retaining the right products when they are young. Their limited ability to shed low-potential products and adopt high-potential ones early in the lifecycle prevents them from operating in product markets where they have strong expertise and constrains performance improvements, leading to potential misallocation of resources.

Literature Review. Our paper is closely related to a line of recent studies on product switching within firms. [Bernard et al. \(2010\)](#) is the closest study to our paper. They use the same dataset as ours (the Census of Manufactures) and study the extent of product switching within firms for the U.S. manufacturing sector. [Broda and Weinstein \(2010\)](#) document the patterns of product entry and exit in consumer goods sectors from

the Nielsen Homescan database. [Bernard and Okubo \(2016\)](#) study the role of product adding and dropping within Japanese manufacturing firms over the business cycle. [Argente et al. \(2018\)](#) use the Nielsen Retail Measurement Services (RMS) scanner data and assess the magnitude of product creation and destruction and product reallocation during and after the Great Recession. [Berlingieri et al. \(2024\)](#) use data on French manufacturing firms and document that rapid, simultaneous additions of many new products play a central role in firm innovation and growth. Other studies show that trade liberalization and competition lead firms to focus on their core products and competence ([Bernard et al., 2011](#); [Eckel and Neary, 2010](#); [Mayer et al., 2014](#)). [Boehm et al. \(2022\)](#) find that firms choose products with common-input capabilities. This paper contributes to this literature by offering a new direct measure of product-specific expertise of firms and establishing novel facts about the product match-quality ladder within firms and its evolution over the firm life-cycle.

This paper also adds to the vast literature on potential factors or frictions that affect the post-entry dynamics and growth of young firms. One strand of studies emphasizes the importance of financing constraints and the role of collateral values for successful young firms ([Evans and Jovanovic, 1989](#); [Holtz-Eakin et al., 1994](#); [Cooley and Quadrini, 2001](#); [Hurst and Lusardi, 2004](#); [Kerr and Nanda, 2009](#); [Robb and Robinson, 2014](#); [Schmalz et al., 2017](#); [Davis and Haltiwanger, 2019](#)). On the other hand, [Foster et al. \(2016\)](#) analyze the process of accumulating demand can create friction for new businesses to grow high. [Akcigit and Ates \(2019\)](#) and [Jo and Kim \(2024\)](#) present barriers to knowledge spillovers as a source to create frictions dampening firm entry and the rapid-growth young firm activities. Furthermore, [Kim \(2025\)](#) demonstrates that uncertain job prospects can pose difficulties to young firms in attracting workers properly and negatively affect firm entry and the growth of high-potential young firms. Along this line, this paper provides a unique channel of product switching to account for young firm growth. In particular, we document new data evidence suggesting that young firms have more difficulties in finding the right product match, and this can give a negative impact on their growth path.

The rest of the paper proceeds as follows. Section 2 presents a theoretical framework of product switching with learning. Section 3 describes data sources and main measures used in our analysis. Section 4 presents our main empirical findings on product dropping and adding activities over the firm lifecycle, the relationship between product dropping and the properties of added products, and the impact of product switching on firm growth, consistent with the model’s predictions. Section 5 discusses additional robustness tests for the main results. Section 6 calibrates the model and conducts counterfactual exercises (in progress). Finally, Section 7 concludes with a discussion on remaining future work.

2 Theoretical Framework

Early in the lifecycle, firms may lack insights into market dynamics and may not fully understand their own potential (e.g., Jovanovic, 1982; Kim, 2025). As a result, they may struggle to assess whether their current products are aligned with consumer demand and perform well in the market, in contrast to mature firms that benefit from greater experience and information. The following section describes a model of product switching over the firm lifecycle with learning that emphasizes the nascency of young firms and their limited information about product markets. The model builds on the learning process in Jovanovic (1982) and product switching behavior of multi-product firms in Bernard et al. (2010). In addition, it incorporates heterogeneity in firms’ information-processing ability, motivated by the data economy literature (e.g., Farboodi et al., 2019; Farboodi and Veldkamp, 2021; Baley and Veldkamp, 2025).

2.1 Environment

The model consists of a finite set of products $p \in \{1, 2, \dots, N\}$ and a continuum of firms (i) operating in each product market. Product markets are monopolistically competitive, with firms producing differentiated varieties within each product. Time is discrete,

denoted by t .

2.2 Consumers

There is a representative household with labor endowment L that supplies labor inelastically and owns all firms. The household chooses consumption to maximize

$$\sum_{t=0}^{\infty} \beta^t \log C_t$$

subject to the period budget constraint

$$\sum_{p=1}^N \int_{i \in \Omega_{pt}} p_{ipt} c_{ipt} di \leq w_t L + \Pi_t,$$

where Ω_{pt} denotes the set of firms operating in product market p , p_{ipt} is the price of good produced by firm i in product p , c_{ipt} is the consumption of it, w_t is the wage, and Π_t is aggregate profits rebated from firms.

The aggregate consumption index C_t is determined by a constant elasticity of substitution (CES) aggregator over a finite set of products $p \in 1, 2, \dots, N$ available in the economy:

$$C_t = \left(\sum_{p=1}^N C_{pt}^{\frac{\nu-1}{\nu}} \right)^{\frac{\nu}{\nu-1}},$$

where the composite goods in product market p , C_{pt} , is given by

$$C_{pt} = \left[\int_{i \in \Omega_p} \theta_{ipt}^{\frac{1}{\eta}} c_{ipt}^{\frac{\eta-1}{\eta}} di \right]^{\frac{\eta}{\eta-1}},$$

which is a CES aggregator over differentiated goods produced by firms operating in market p . The term θ_{ipt} is a product-specific demand shifter that captures how well firm i 's product performs in the market.⁴ The parameters ν and η denote the constant

⁴ θ_{ipt} captures firm-product-specific fundamentals that determine market performance, such as how appealing the product is to consumers and how strongly it aligns with demand conditions. We refer to

elasticities of substitution across products and across goods within each product market, respectively, with $1 < \nu < \eta$.

The household minimizes the cost of attaining C_t given the budget constraint, which yields the following demand function for each good:⁵

$$c_{ipt} = P_{pt}^\eta C_{pt} \theta_{ipt} p_{ipt}^{-\eta}. \quad (2.1)$$

The aggregate price index associated with the composite good C_t is

$$P_t = \left(\sum_{p=1}^N P_{pt}^{1-\nu} \right)^{\frac{1}{1-\nu}}, \quad (2.2)$$

where the product-level price index is given by

$$P_{pt} = \left(\int_{i \in \Omega_p} \theta_{ipt} p_{ipt}^{1-\eta} di \right)^{\frac{1}{1-\eta}}. \quad (2.3)$$

2.3 Firms

Firms are born with a fundamental type φ_i , which determines productivity $z_i \equiv z(\varphi_i)$ in the production process, capacity $n_i \equiv n(\varphi_i)$, which is the maximum number of product lines that the firm can operate at a time and determines the scope of the firm, and managerial practices $m_i \equiv m(\varphi_i)$. The mappings satisfy $z : \mathbb{R} \rightarrow \mathbb{R}$ with $z'(\cdot) > 0$, $n : \mathbb{R} \rightarrow \mathbb{N}$ with $N'(\cdot) > 0$, and $m : \mathbb{R} \rightarrow \mathbb{R}$ with $m'(\cdot) > 0$. In each period, firms decide for each product line whether to add or drop a product and choose quantities for all operating products.

this as the firm's product-specific expertise.

⁵The corresponding market-clearing price is $p_{ipt}^* = P_{pt} C_{pt}^{\frac{1}{\eta}} \theta_{ipt}^{\frac{1}{\eta}} c_{ipt}^{-\frac{1}{\eta}}$.

2.3.1 Firm-Product Match and Learning

In addition to their fundamental type, firms (i) are born with time-invariant, product-specific expertise θ_{ip} for each product p . θ_{ip} is drawn from a log-normal distribution, $\log \theta_{ip} \sim N(\theta_0, \sigma_0^2)$. A higher value of θ_{ip} indicates a better match between the firm and the product, reflecting greater expertise in operating that product, and thus leading to higher profitability. We assume that this expertise is embodied in product quality and enters the demand function.⁶

Firms do not observe θ_{ip} directly. Instead, in each period they observe a noisy signal inferred from the market-clearing price induced by realized demand:

$$\log \theta_{ipt} = \log \theta_{ip} + \varepsilon_{ipt},$$

where $\varepsilon_{ipt} \sim N(0, \sigma_\varepsilon^2)$ is an i.i.d. shock that follows a normal distribution. We assume that learning follows Bayesian updating based on $\log \theta_{ipt}$ and an additional signal $\log s_{ipt}$ observed each period (Farboodi et al., 2019; Farboodi and Veldkamp, 2021; Baley and Veldkamp, 2025), where the signal is extracted using the firm's information-processing ability ψ_{it} :

$$\log s_{ipt} = \log \theta_{ip} + v_{ipt}, \quad v_{ipt} \sim N(0, \frac{\sigma_\varepsilon^2}{\psi_{it}}). \quad (2.4)$$

A firm's ability to process observable information depends on its managerial practices m_{it} (a function of φ_i and, potentially, training), its experience (captured by firm age, a_{it}), and the size of its accumulated data D_{it} (which depends on a_{it} , product scope n_i , and potentially other data sources). We summarize these factors by $\psi_{it} \equiv \psi(a_{it}, m_{it}, D_{it}; \varphi_i)$, with $\partial \psi_{it} / \partial x \geq 0, \forall x \in \{a_{it}, m_{it}, D_{it}, \varphi_i\}$.⁷ This ability increases the precision of the signals firms draw. If managerial training and other data sources are exogenously given and fixed, then firm age (a_{it}) fully determines the evolution of ψ_{it} .⁸

⁶Alternatively, it can be modeled on the production side as product-level productivity.

⁷The cumulative number of products dropped can serve as an additional proxy for firm experience. We abstract from this variable because it introduces dependencies across product lines, which substantially reduce the tractability of the model.

⁸Then, the state variables that need to be tracked reduce to $(\hat{\theta}_{ipt}, \tau_{ipt}; a_{it}, \varphi_i)$.

Given this structure, learning proceeds as follows. Firm i enters product market p with an initial belief about θ_{ip} , given by the prior $\log \theta_{ip} \sim N(\theta_0, \sigma_0^2)$. After production and sales, the firm observes realized outcome and updates its beliefs each period. In period t , conditional on having operated in product market p for τ periods (firm-product tenure) and observing the contemporaneous outcome θ_{ipt} , its information set is

$$\mathcal{I}_{ipt} \equiv \{\theta_{ip\tilde{t}}, s_{ip\tilde{t}}\}_{\tilde{t}=t-\tau+1}^t. \quad (2.5)$$

This yields the posterior belief:

$$\hat{\theta}_{ipt} = \hat{\sigma}_{ipt}^2 \left(\frac{\theta_0}{\sigma_0^2} + \frac{\tau}{\sigma_\varepsilon^2} \frac{\sum_{\tilde{t}=t-\tau+1}^t \log \theta_{ip\tilde{t}}}{\tau} + \frac{\sum_{\tilde{t}=t-\tau+1}^t \psi_{i\tilde{t}} \log s_{ip\tilde{t}}}{\sigma_\varepsilon^2} \right) \quad (2.6)$$

$$\hat{\sigma}_{ipt}^2 = \left(\frac{1}{\sigma_0^2} + \frac{\tau}{\sigma_\varepsilon^2} + \frac{\sum_{\tilde{t}=t-\tau+1}^t \psi_{i\tilde{t}}}{\sigma_\varepsilon^2} \right)^{-1}, \quad (2.7)$$

where $\hat{\theta}_{ipt}$ and $\hat{\sigma}_{ipt}^2$ is the mean and variance of the firm's posterior belief about $\log \theta_{ip}$, respectively.

Importantly, the posterior mean is a weighted average of the prior θ_0 , the average observed outcome $\frac{1}{\tau} \sum_{\tilde{t}=t-\tau+1}^t \log \theta_{ip\tilde{t}}$, and signals $\{\log s_{ip\tilde{t}}\}_{\tilde{t}=t-\tau+1}^t$, with weights determined by tenure τ and the firm's time-varying information-processing ability $\{\psi_{i\tilde{t}}\}_{\tilde{t}=t-\tau+1}^t$. This structure allows us to identify multiple sources of learning and to disentangle the role of the firm's information-processing ability (which evolves with firm age) from the effect of product-specific tenure. Given this, we can characterize the posterior mean and variance of θ_{ip} , as well as beliefs about future realizations θ_{ipt+1} . Since the posterior variance is fully determined by product-firm tenure, firm age, information-processing ability, and firm type, posterior beliefs can be tracked by $(\hat{\theta}_{ipt}, \tau; a_{it}, \psi_{it}, \varphi_i)$. All firms start with a common initial prior $(\hat{\theta}_{ip}^0, 0; a_{it}, \psi_{it}, \varphi_i)$, corresponding to $(\hat{\theta}_{ip}^0, \hat{\sigma}_{ip}^0) \equiv (\theta_0, \sigma_0)$.

Based on this, we characterize how posterior beliefs about product-specific expertise θ_{ip} vary with firms' information-processing ability over the lifecycle:

Lemma 1 (Biased Learning). *Holding all else equal, lower information-processing ability*

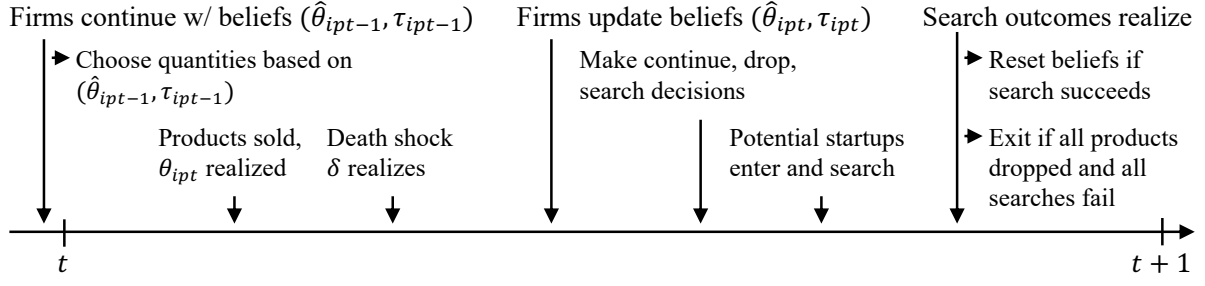


Figure 1: Timeline of the model

ψ_{it} induces biased beliefs: underestimation of $\hat{\theta}_{ipt}$ for high θ_{ip} and overestimation for low θ_{ip} , along with higher posterior variance.

Proof. See Appendix A.1. □

Since ψ_{it} is closely related to firm age a_{it} , Lemma 1 implies that firms exhibit systematically biased posterior beliefs when they are young. Lower information-processing ability makes their signals noisier and less informative, so posterior beliefs place greater weight on the prior θ_0 . As a result, when product-specific expertise is high ($\log \theta_{ip} > \theta_0$), young firms place insufficient weight on signals and underestimate $\hat{\theta}_{ipt}$, whereas when it is low ($\log \theta_{ip} < \theta_0$), they overestimate it.

Noisier signals also imply less precise posteriors. As ψ_{it} increases with age, signals become more informative and firms place greater weight on observed outcomes, which reduces posterior variance. Consequently, when firms are young, they hold both more biased and more diffuse beliefs, which can lead to systematically different product-switching decisions early in the lifecycle.

2.3.2 Timing of the Model

The timing of the model is illustrated in figure 1. At the beginning of period t , firms enter with initial posterior beliefs about θ_{ip} and predetermined quantities, both based on $\mathcal{I}_{ipt-1}$, prior to the realization of θ_{ipt} .⁹ Demand and sales are then realized given

⁹For entrant firms, the initial posterior coincides with the initial prior $(\hat{\theta}_{ip}^0, \hat{\sigma}_{ip}^0)$.

θ_{ipt} , after which the death shock δ is realized. Surviving firms update their posterior beliefs based on realized outcomes and decide whether to continue operating or drop the product, and whether to switch to a new product via search.

Potential startups also conduct searches to enter product markets. Search outcomes are realized at the end of the period. A firm exits the economy if it drops all of its products and fails all of its searches. Firms that continue into the next period choose quantities based on updated beliefs, where firms that switch and successfully enter a new product market reset their beliefs to the initial prior $(\hat{\theta}_{ipt}^0, \hat{\sigma}_{ipt}^0)$. Accordingly, we define firm decisions and value function after θ_{ipt} is realized and beliefs are updated, when the search decision is made.

2.3.3 Production

The firm produces each product using linear production technology with labor as the only input, $y_{ipt} = z_i l_{ipt}$. Firms choose quantity based on their posterior beliefs about θ_{ip} before observing the contemporaneous realization θ_{ipt} in each period. Production incurs an operating fixed cost $\mathcal{F}^o$ for each product.

For continuing products at $t + 1$, the firm chooses output y_{ipt+1} to maximize expected per-period profit $\mathbb{E}_{\theta_{ipt+1}} \pi_{it+1}(\hat{\theta}_{ipt}, \tau_{ipt}) \equiv \mathbb{E}_{\theta_{ipt+1}} \pi(\hat{\theta}_{ipt}, \tau_{ipt}; a_{it}, \psi_{it}, \varphi_i)$ with $z_i = z(\varphi_i)$, where expectations are taken over θ_{ipt+1} conditional on the firm's posterior beliefs $(\hat{\theta}_{ipt}, \tau_{ipt}; a_{it}, \psi_{it}, \varphi_i)$:

$$\mathbb{E}_{\theta_{ipt+1}} \pi_{it+1}(\hat{\theta}_{ipt}, \tau_{ipt}) \equiv \max_{y_{ipt+1}} \mathbb{E}_{\theta_{ipt+1}} \left[p_{ipt+1}(\theta_{ipt+1}) | (\hat{\theta}_{ipt}, \tau_{ipt}; a_{it}, \psi_{it}, \varphi_i) \right] y_{ipt+1} - \frac{w_{t+1}}{z_i} y_{ipt+1} - \mathcal{F}^o \quad (2.8)$$

subject to the demand function (2.1). The optimal quantity choice is

$$y_{ipt+1}^* = \left(\frac{\eta - 1}{\eta} \right)^\eta P_{pt+1}^\eta C_{pt+1} \left[\mathcal{B}_{it}(\hat{\theta}_{ipt}, \tau_{ipt}) \right]^\eta z_i^\eta w_{t+1}^{-\eta}, \quad (2.9)$$

where $\mathcal{B}_{it}(\hat{\theta}_{ipt}, \tau_{ipt}) \equiv \mathcal{B}(\hat{\theta}_{ipt}, \tau_{ipt}; a_{it}, \psi_{it}, \varphi_i)$ denotes the posterior mean of $\theta_{ipt+1}^{1/\eta}$ given

$(\hat{\theta}_{ipt}, \tau_{ipt})$:

$$\mathcal{B}_{it}(\hat{\theta}_{ipt}, \tau_{ipt}) \equiv \mathbb{E}_{\theta_{ipt+1}} \left[\theta_{ipt+1}^{1/\eta} | (\hat{\theta}_{ipt}, \tau_{ipt}; a_{it}, \psi_{it}, \varphi_i) \right] = \exp \left\{ \frac{\hat{\theta}_{ipt}}{\eta} + \frac{1}{2} \left(\frac{\hat{\sigma}_{ipt}^2 + \sigma_\varepsilon^2}{\eta^2} \right) \right\}. \quad (2.10)$$

Substituting this optimal quantity into (2.1), the market-clearing price, conditional on the realized θ_{ipt} is

$$p_{ipt+1} = \frac{\eta}{\eta - 1} \frac{w_{t+1}}{z_i} \frac{\theta_{ipt+1}^{1/\eta}}{\mathcal{B}_{it}(\hat{\theta}_{ipt}, \tau_{ipt})}. \quad (2.11)$$

Thus, prior to observing θ_{ipt+1} , the firm's expected price is the standard constant markup over marginal cost:

$$\mathbb{E}_{\theta_{ipt+1}} \left[p_{ipt+1} | (\hat{\theta}_{ipt}, \tau_{ipt}; a_{it}, \psi_{it}, \varphi_i) \right] = \frac{\eta}{\eta - 1} \frac{w_{t+1}}{z_i}. \quad (2.12)$$

Substituting (2.12) into (2.8), the optimal expected per-period profit in product market p is given by

$$\begin{aligned} \pi_{it+1}^*(\hat{\theta}_{ipt}, \tau_{ipt}) &\equiv \mathbb{E}_{\theta_{ipt+1}} \pi_{it+1}(\hat{\theta}_{ipt}, \tau_{ipt}) \\ &= \frac{(\eta - 1)^{\eta-1}}{\eta^\eta} \left(\frac{z_i}{w_{t+1}} \right)^{\eta-1} P_{pt+1}^\eta C_{pt+1} \left[\mathcal{B}_{it}(\hat{\theta}_{ipt}, \tau_{ipt}) \right]^\eta - \mathcal{F}^o. \end{aligned} \quad (2.13)$$

2.3.4 Product Switching and Exit

Each firm i is endowed with a fixed number of product lines $k \in \{1, 2, \dots, n_i\}$, which represent capacity slots that can be allocated to different products over time. At any point in time, each product line k is assigned to at most one product $p \in \{1, \dots, N\} \cup \emptyset$, where $\emptyset$ denotes a vacant line. The mapping from product lines to products is endogenous and evolves over time, where the destination product market is determined randomly. In particular, a firm can reassign a product line to a different product through switching.¹⁰

¹⁰For example, a line initially assigned to product $p = 1$ can be reassigned to product $p = 3$ in a subsequent period. This distinction implies that product lines are the units of capacity and decision-making, while product indices represent the set of markets in which firms can operate.

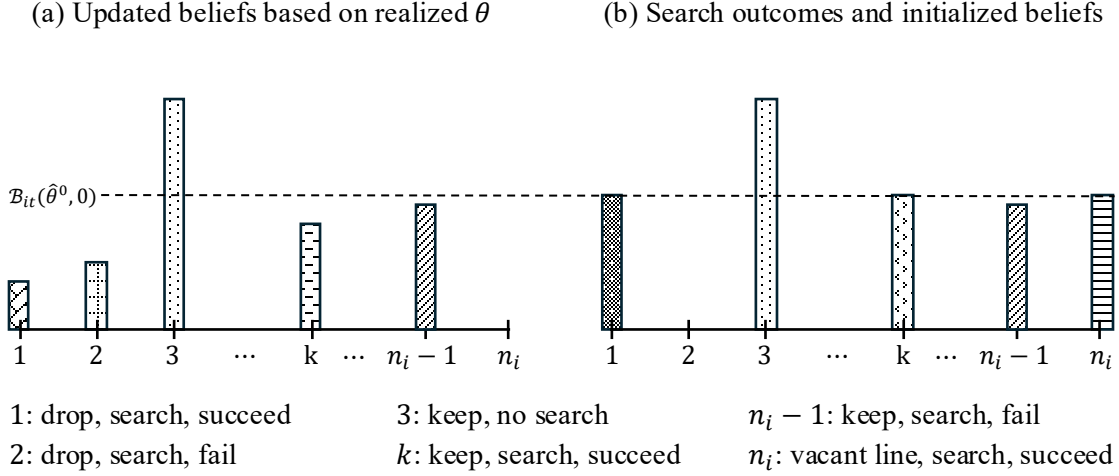


Figure 2: Evolution of product lines for firm i with scope $n_i = n(\varphi_i)$

For each product line, the firm can search for a new product and switch to it by paying a search cost $\mathcal{F}^{ss}$. Search succeeds with probability x . Given the two fixed costs $\mathcal{F}^o$ and $\mathcal{F}^{ss}$, there exist two cutoffs for $\mathcal{B}_{it}(\hat{\theta}_{ipt}, \tau_{ipt})$: one for search, denoted by $\mathcal{B}_{it}^{ss}(\hat{\theta}_{ipt}, \tau_{ipt}) \equiv \mathcal{B}^{ss}(\hat{\theta}_{ipt}, \tau_{ipt}; a_{it}, \psi_{it}, \varphi_i)$, and another for product dropping, denoted by $\mathcal{B}_{it}^d(\hat{\theta}_{ipt}, \tau_{ipt}) \equiv \mathcal{B}^d(\hat{\theta}_{ipt}, \tau_{ipt}; a_{it}, \psi_{it}, \varphi_i)$. For each product, if $\mathcal{B}_{it}(\hat{\theta}_{ipt}, \tau_{ipt})$ is below the search threshold $\mathcal{B}_{it}^{ss}(\hat{\theta}_{ipt}, \tau_{ipt})$, the firm incurs a search cost and replaces the focal product upon a successful match. If the focal product line is vacant and the firm does not exit, it initiates a search. If $\mathcal{B}_{it}(\hat{\theta}_{ipt}, \tau_{ipt})$ is below the drop threshold $\mathcal{B}_{it}^d(\hat{\theta}_{ipt}, \tau_{ipt})$ in the current period, the firm drops the product but still initiates a search. The firm exits the economy if all products lie in the range of $\mathcal{B}_{it}(\hat{\theta}_{ipt}, \tau_{ipt}) < \mathcal{B}_{it}^d(\hat{\theta}_{ipt}, \tau_{ipt})$ and all searches fail, or exogenously with probability δ .¹¹

Figure 2 illustrates possible outcomes for firm i with product scope $n_i = n(\varphi_i)$ before and after product-switching decisions. Panel (a) shows initial $\mathcal{B}_{it}(\hat{\theta}, \tau)$ for each product line, and Panel (b) shows the resulting outcomes. The height of each bar represents $\mathcal{B}_{it}(\hat{\theta}, \tau)$. Each product line is occupied by a distinct product, indicated by different patterns.

¹¹Here, we assume that the two thresholds are functions of $(\hat{\theta}_{ipt}, \tau_{ipt}; a_{it}, \psi_{it}, \varphi_i)$. This reflects that the value of $\mathcal{B}_{it}(\hat{\theta}_{ipt}, \tau_{ipt})$ depends on both the level of the posterior mean and the standard deviation, and that ψ_{it} affects the evolution of these two components differently. We establish this result below.

Products in lines 1 and 2 have $\mathcal{B}_{it}(\hat{\theta}, \tau)$ below the drop thresholds, so the firm drops them and initiates searches. The product in line 3 has $\mathcal{B}_{it}(\hat{\theta}, \tau)$ above the search threshold, so the firm continues without searching. Products in lines k and $n_i - 1$ have $\mathcal{B}_{it}(\hat{\theta}, \tau)$ between the drop cutoff and the search threshold; the firm keeps these products but initiates searches. Line n_i is vacant, so the firm searches to fill it.

Searches succeed for lines 1, k , and n_i , but fail for lines 2 and $n_i - 1$. As a result, the posterior beliefs in lines 1, k , and n_i are reset to $\mathcal{B}_{it}(\hat{\theta}^0, 0)$, corresponding to the initial prior belief. Line 2 becomes vacant, while line n_i is filled with a new product. An additional case arises when a product line starts vacant and remains vacant due to search failure.

2.3.5 Value Function of Product Lines

Since product-switching decisions are made at the product-line level and some lines may be vacant, we formulate firm decisions and value functions at the product-line level. However, the endogenous exit decision depends on the composition of the firm's portfolio of posterior beliefs about θ_{ip} , so the product-line value function generally depends on the firm-level product portfolio. In the next section, we show that, under a plausible assumption, this linkage can be broken, allowing product-line decisions to be analyzed independently of the portfolio.

Let $V_{it}^{(\cdot)}(\hat{\theta}_{ipkt}, \tau_{ipkt}) \equiv V^{(\cdot)}(\hat{\theta}_{ipkt}, \tau_{ipkt}; a_{it}, \psi_{it}, \varphi_i)$ for notational simplicity. The value function for product line k is given by

$$V_{it}(\hat{\theta}_{ipkt}, \tau_{ipkt}) = \max\{V_{it}^{ns}(\hat{\theta}_{ipkt}, \tau_{ipkt}), V_{it}^{ss}(\hat{\theta}_{ipkt}, \tau_{ipkt})\}, \quad (2.14)$$

where $V_{it}^{ns}(\hat{\theta}_{ipkt}, \tau_{ipkt})$ is the value without search, and $V_{it}^{ss}(\hat{\theta}_{ipkt}, \tau_{ipkt})$ is the value with search, which includes the options to drop the product, search for a new one, or firm exit. This gives rise to two cutoffs: a search cutoff, $\mathcal{B}_{it}^{ss}(\hat{\theta}_{ipt}, \tau_{ipt})$, and a product drop cutoff, $\mathcal{B}_{it}^d(\hat{\theta}_{ipt}, \tau_{ipt})$.¹² Given the cutoffs, we can characterize each value function for

¹²Note that there is no separate search cutoff for dropped or vacant products, as we assume the value

each case.

(a) No search ($\mathcal{B}_{it}(\hat{\theta}_{ip_k t}, \tau_{ip_k t}) \geq \mathcal{B}_{it}^{ss}(\hat{\theta}_{ip_k t}, \tau_{ip_k t})$): the firm does not conduct search in the current period. However, as $\theta_{ip_k t}$ evolves and posterior beliefs are updated, the firm may initiate search in subsequent periods.

$$V_{it}^{ns}(\hat{\theta}_{ip_k t}, \tau_{ip_k t}) = \beta \left[\pi_{it+1}^*(\hat{\theta}_{ip_k t}, \tau_{ip_k t}) + \mathbb{E}_{(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}) | (\hat{\theta}_{ip_k t}, \tau_{ip_k t})} V_{it+1}(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}) \right],$$

where the last term,

$$\begin{aligned} & \mathbb{E}_{(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}) | (\hat{\theta}_{ip_k t}, \tau_{ip_k t})} V_{it+1}(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}) = \\ & (1 - \delta) \left[\int_{\mathcal{R}_{it+1}^{ns}} V_{it+1}^{ns}(\hat{\theta}', \tau') g_{it}(\hat{\theta}', \tau' | \hat{\theta}_{ip_k t}, \tau_{ip_k t}) d\hat{\theta}' + \int_{\mathcal{R}_{it+1}^{ss}} V_{it+1}^{ss}(\hat{\theta}', \tau') g_{it}(\hat{\theta}', \tau' | \hat{\theta}_{ip_k t}, \tau_{ip_k t}) d\hat{\theta}' \right. \\ & \left. + \int_{\mathcal{R}_{it+1}^d} V_{it+1}^{ss}(\mathbf{0}) g_{it}(\hat{\theta}', \tau' | \hat{\theta}_{ip_k t}, \tau_{ip_k t}) d\hat{\theta}' \right] + \delta V_{it+1}^\delta, \end{aligned}$$

is the expected continuation value of product line k in the absence of product switching, with posterior beliefs $(\hat{\theta}_{ip_k t}, \tau_{ip_k t})$ evolving over time. Here, $\tau' = \tau_{ipt} + 1$, and $g_{it}(\cdot | \hat{\theta}_{ipt}, \tau_{ipt}) \equiv g(\cdot | \hat{\theta}_{ipt}, \tau_{ipt}; a_{it}, \psi_{it}, \varphi_i)$ denotes the transition probability of posterior beliefs induced by Bayesian updating in (2.6) and (2.7). The regions are defined by decision rules in posterior space: $\mathcal{R}_{it+1}^{ns} = \{(\hat{\theta}', \tau') : \mathcal{B}(\hat{\theta}', \tau') \geq \mathcal{B}_{it+1}^{ss}(\hat{\theta}', \tau')\}$, $\mathcal{R}_{it+1}^{ss} = \{(\hat{\theta}', \tau') : \mathcal{B}_{it+1}^d(\hat{\theta}', \tau') \leq \mathcal{B}(\hat{\theta}', \tau') < \mathcal{B}_{it+1}^{ss}(\hat{\theta}', \tau')\}$, and $\mathcal{R}_{it+1}^d = \{(\hat{\theta}', \tau') : \mathcal{B}(\hat{\theta}', \tau') < \mathcal{B}_{it+1}^d(\hat{\theta}', \tau')\}$. The third term on the RHS includes the firm's endogenous exit case. V_{it+1}^δ denotes the firm's exit value per product line under exogenous exit. We assume $V_{it+1}^\delta = V_{it+1}^{ss}(\mathbf{0})$. We discuss this in detail below.

(b) Search with switching ($\mathcal{B}_{it}(\hat{\theta}_{ip_k t}, \tau_{ip_k t}) < \mathcal{B}_{it}^{ss}(\hat{\theta}_{ip_k t}, \tau_{ip_k t})$): In this region, the firm searches for a new product in the current period, either to replace its continuing product or to fill a vacant product line. More specifically, if $\mathcal{B}_{it}^d(\hat{\theta}_{ip_k t}, \tau_{ip_k t}) \leq \mathcal{B}_{it}(\hat{\theta}_{ip_k t}, \tau_{ip_k t}) < \mathcal{B}_{it}^{ss}(\hat{\theta}_{ip_k t}, \tau_{ip_k t})$, the firm continues operating the current product but replaces it with

of search is non-negative and firms always search to fill unoccupied product lines.

a newly obtained product if the search succeeds at the end of the current period. If $\mathcal{B}_{it}(\hat{\theta}_{ip_k t}, \tau_{ip_k t}) < \mathcal{B}_{it}^d(\hat{\theta}_{ip_k t}, \tau_{ip_k t})$, the firm drops the product regardless of the search outcome and fills the vacant slot only if search is successful; otherwise, the slot remains vacant.

In the first case, the value of search with switching is

$$\begin{aligned} V_{it}^{ss}(\hat{\theta}_{ip_k t}, \tau_{ip_k t}) = & \\ & - \mathcal{F}^{ss} + x\beta \left[\pi_{it+1}^*(\hat{\theta}_{ip'_k t}, \tau_{ip'_k t}^0) + \mathbb{E}_{(\hat{\theta}_{ip'_k t+1}, \tau_{ip'_k t+1}) | (\hat{\theta}_{ip'_k t}, \tau_{ip'_k t}^0)} V_{it+1}(\hat{\theta}_{ip'_k t+1}, \tau_{ip'_k t+1}) \right] \\ & + (1-x)\beta \left[\pi_{it+1}^*(\hat{\theta}_{ip_k t}, \tau_{ip_k t}) + \mathbb{E}_{(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}) | (\hat{\theta}_{ip_k t}, \tau_{ip_k t})} V_{it+1}(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}) \right]. \end{aligned}$$

When search succeeds with probability x and the product is switched, the firm's expertise in the new product market p' for product line k , $\theta_{ip'_k}$, is drawn from the initial prior distribution. Accordingly, the firm's posterior beliefs are reset to the initial posterior $(\hat{\theta}_{ip'_k t}, \tau_{ip'_k t}^0)$, where $\tau_{ip'_k t}^0 = 0$.

The search is random and assumed in the following way. For each firm, define the exclusion set as the union of currently operated products and the most recent product associated with each vacant line, which has cardinality $n(\varphi_i)$. Hence, all product lines search over a common feasible set of size $N - n(\varphi_i)$, with assignments coordinated without replacement. For a vacant line, if search fails, the associated product is updated to the most recent product reached but not entered.¹³ Under uniform draws and symmetric product markets, expectations over product draws collapse and therefore do not appear explicitly in the value function. When search fails with probability $1 - x$, the value function coincides with the no-search case, as the firm continues to operate the existing product.

In the second case, where $\mathcal{B}_{it}(\hat{\theta}_{ip_k t}, \tau_{ip_k t}) < \mathcal{B}_{it}^d(\hat{\theta}_{ip_k t}, \tau_{ip_k t})$, the value of search with

¹³We assume that when a firm conducts a search, it first draws a product uniformly from the feasible set, after which entry success is determined.

switching is:

$$V_{it}^{ss}(\mathbf{0}) = -\mathcal{F}^{ss} + x\beta \left[\pi_{it+1}^*(\hat{\theta}_{ip'_k t}^0, \tau_{ip'_k t}^0) + \mathbb{E}_{(\hat{\theta}_{ip'_k t+1}, \tau_{ip'_k t+1}) | (\hat{\theta}_{ip'_k t}^0, \tau_{ip'_k t}^0)} V_{it+1}(\hat{\theta}_{ip'_k t+1}, \tau_{ip'_k t+1}) \right] \\ + (1-x)\beta V_{it+1}^{ss}(\mathbf{0}),$$

where the focal product line remains vacant, and the continuation value after failed search is $V_{i,t+1}^{ss}(\mathbf{0})$.¹⁴ We assume that the firm's exit value per product line equals the value of a vacant line when both are evaluated simultaneously.¹⁵ In this case, it equals the value of waiting until $t+1$ and conducting a search, as the firm exits only when all product lines are dropped and all searches fail.¹⁶

$$V_{it}^\emptyset = \beta V_{it+1}^{ss}(\mathbf{0}).$$

With this, we can characterize the search and product-dropping cutoffs, $\mathcal{B}_{it}^{ss}(\hat{\theta}_{ipt}, \tau_{ipt})$ and $\mathcal{B}_{it}^d(\hat{\theta}_{ipt}, \tau_{ipt})$. First, with posterior beliefs evaluated at the search cutoff, $\mathcal{B}_{it}(\hat{\theta}_{ip_k t}^{ss}, \tau_{ip_k t}^{ss}) = \mathcal{B}_{it}^{ss}(\hat{\theta}_{ip_k t}^{ss}, \tau_{ip_k t}^{ss})$, firms are indifferent between searching and not searching, which yields the following condition:

$$V_{it}^{ns}(\hat{\theta}_{ip_k t}^{ss}, \tau_{ip_k t}^{ss}) = V_{it}^{ss}(\hat{\theta}_{ip_k t}^{ss}, \tau_{ip_k t}^{ss}) \\ \Rightarrow V_{it}^{ns}(\hat{\theta}_{ip'_k t}^0, \tau_{ip'_k t}^0) - V_{it}^{ns}(\hat{\theta}_{ip_k t}^{ss}, \tau_{ip_k t}^{ss}) = \frac{\mathcal{F}^{ss}}{x},$$

where $V_{it}^{ns}(\hat{\theta}_{ip'_k t}^0, \tau_{ip'_k t}^0)$ is the term multiplied by x in $V_{it}^{ss}(\hat{\theta}_{ip_k t}^{ss}, \tau_{ip_k t}^{ss})$. Thus, $(\hat{\theta}_{ip_k t}^{ss}, \tau_{ip_k t}^{ss})$ is

¹⁴The initial prior beliefs $(\hat{\theta}^0, \tau^0)$ are common across firms and invariant over time, and $\pi_i^*(\hat{\theta}^0, \tau^0)$ is likewise time-invariant. In addition, $\lim_{\tau \rightarrow \infty} ((1-x)\beta)^{\tau+1} V_{i,t+\tau}^{ss}(\mathbf{0}) = 0$. Thus, iterating forward yields

$$V_{it}^{ss}(\mathbf{0}) = \frac{x\beta\pi_{it+1}^*(\hat{\theta}^0, \tau^0) - \mathcal{F}^{ss}}{1 - (1-x)\beta} + x\beta \sum_{\tau=0}^{\infty} \mathbb{E}_{(\hat{\theta}', \tau') | (\hat{\theta}^0, \tau^0)} [(1-x)\beta]^\tau V_{it+1+\tau}(\hat{\theta}', \tau').$$

¹⁵Thus, in the case of exogenous exit, $V_{it+1}^\delta = V_{it+1}^{ss}(\mathbf{0})$.

¹⁶This assumption is made for tractability. Under this assumption, the term following $(1-x)$ in $V_{it}^{ss}(\mathbf{0})$ simplifies to $(1 - \text{firm exit probability}) \times \beta V_{it+1}^{ss}(\mathbf{0}) + (\text{firm exit probability}) \times \beta V_{it+1}^{ss}(\mathbf{0}) = \beta V_{it+1}^{ss}(\mathbf{0})$, so the exit probability drops out of the value function.

the belief that makes the difference between the continuation value of a newly drawn product under the initial posterior and the continuation value under the current posterior equal to the search cost scaled by the success probability.

In a similar fashion, evaluating posterior beliefs at the product-dropping cutoff, $(\hat{\theta}_{ip_k t}^d, \tau_{ip_k t}^d)$ satisfying $\mathcal{B}(\hat{\theta}_{ip_k t}^d, \tau_{ip_k t}^d) = \mathcal{B}_{it}^d(\hat{\theta}_{ip_k t}^d, \tau_{ip_k t}^d)$, firms are indifferent between search with product switching and product dropping. This gives the following condition:

$$V_{it}^{ss}(\hat{\theta}_{ip_k t}^d, \tau_{ip_k t}^d) = V_{it}^{ss}(\mathbf{0}).$$

If the firm fails to search in a vacant product line but does not exit, it initiates a new search in the following period. Since

$$\begin{aligned} V_{it}^{ss}(\hat{\theta}_{ip_k t}^d, \tau_{ip_k t}^d) = & -\mathcal{F}^{ss} + x\beta \left[\pi_{it+1}^*(\hat{\theta}_{ip_k' t}^0, \tau_{ip_k' t}^0) + \mathbb{E}_{(\hat{\theta}_{ip_k' t+1}, \tau_{ip_k' t+1}) | (\hat{\theta}_{ip_k' t}^0, \tau_{ip_k' t}^0)} V_{it+1}(\hat{\theta}_{ip_k' t+1}, \tau_{ip_k' t+1}) \right] \\ & + (1-x)\beta \left[\pi_{it+1}^*(\hat{\theta}_{ip_k t}^d, \tau_{ip_k t}^d) + \mathbb{E}_{(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}) | (\hat{\theta}_{ip_k t}^d, \tau_{ip_k t}^d)} V_{it+1}(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}) \right], \end{aligned}$$

it follows that¹⁷

$$\pi_{it+1}^*(\hat{\theta}_{ip_k t}^d, \tau_{ip_k t}^d) + \mathbb{E}_{(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}) | (\hat{\theta}_{ip_k t}^d, \tau_{ip_k t}^d)} V_{it+1}(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}) = V_{it+1}^{ss}(\mathbf{0}).$$

Lastly, the product-line value function can be fully characterized over the state space $(\hat{\theta}_{ipt}, \tau_{ipt}; a_{it}, \varphi_i)$ for the following reason. Under our functional assumption, ψ_{it} is fully determined by (a_{it}, φ_i) . Under non-switching, tenure evolves deterministically. Thus, the transition distribution is degenerate in τ and Gaussian in $\hat{\theta}$. At time t , conditional on current information, $\log \theta_{ip} \mid (\hat{\theta}_{ipt}, \tau_{ipt}, a_{it}, \varphi_i) \sim N(\hat{\theta}_{ipt}, \hat{\sigma}_{ipt}^2)$, where $\hat{\sigma}_{ipt}^2$ is determined by τ_{ipt} , firm age, and firm type. In period $t+1$, the firm observes $\log \theta_{ipt+1} = \log \theta_{ip} + \varepsilon_{ipt+1}$, where $\varepsilon_{ipt+1} \sim N(0, \sigma_\varepsilon^2)$, and $\log s_{ipt+1} = \log \theta_{ip} + v_{ipt+1}$, where

¹⁷Note that

$$V_{it+1}^{ss}(\mathbf{0}) = \frac{x\beta\pi_{it+1}^*(\hat{\theta}^0, \tau^0) - \mathcal{F}^{ss}}{1 - (1-x)\beta} + x\beta \sum_{\tau=0}^{\infty} \mathbb{E}_{(\hat{\theta}', \tau') | (\hat{\theta}^0, \tau^0)} [(1-x)\beta]^\tau V_{it+2+\tau}(\hat{\theta}', \tau').$$

$v_{ipt+1} \sim N\left(0, \frac{\sigma_\varepsilon^2}{\psi_{it+1}}\right)$. Bayesian updating then implies $\hat{\sigma}_{ipt+1}^2 = \left(\frac{1}{\hat{\sigma}_{ipt}^2} + \frac{1+\psi_{it+1}}{\sigma_\varepsilon^2}\right)^{-1}$, and $\hat{\theta}_{ipt+1} = \hat{\sigma}_{ipt+1}^2 \left(\frac{\hat{\theta}_{ipt}}{\hat{\sigma}_{ipt}^2} + \frac{\log \theta_{ipt+1}}{\sigma_\varepsilon^2} + \frac{\psi_{it+1} \log s_{ipt+1}}{\sigma_\varepsilon^2}\right)$, which characterize the posterior belief about $\log \theta_{ip}$.

By substituting $\log \theta_{ipt+1}$ and $\log s_{ipt+1}$, $\hat{\theta}_{ipt+1}$ can be rewritten as

$$\hat{\theta}_{ipt+1} = \underbrace{\hat{\sigma}_{ipt+1}^2 \left(\frac{\hat{\theta}_{ipt}}{\hat{\sigma}_{ipt}^2} + \frac{(1 + \psi_{it+1}) \log \theta_{ip}}{\sigma_\varepsilon^2} \right)}_{\text{depends on } \log \theta_{ip}} + \underbrace{\hat{\sigma}_{ipt+1}^2 \frac{\varepsilon_{ipt+1} + \psi_{it+1} v_{ipt+1}}{\sigma_\varepsilon^2}}_{\text{noise}}.$$

Integrating out $\log \theta_{ip}$, ε_{ipt+1} , and v_{ipt+1} conditional on the current posterior yields $\hat{\theta}_{ipt+1} \mid (\hat{\theta}_{ipt}, \tau_{ipt}) \sim N\left(\hat{\theta}_{ipt}, \hat{\sigma}_{ipt}^2 - \hat{\sigma}_{ip,t+1}^2\right)$, and $\tau_{ip,t+1} = \tau_{ipt} + 1$.¹⁸ Therefore, the transition probability has the form

$$g_{it}(\hat{\theta}_{ipt+1}, \tau_{ipt+1} \mid \hat{\theta}_{ipt}, \tau_{ipt}) = \phi\left(\hat{\theta}_{ipt+1}; \hat{\theta}_{ipt}, \hat{\sigma}_{ipt}^2 - \hat{\sigma}_{ip,t+1}^2\right) \mathbf{1}\{\tau_{ipt+1} = \tau_{ipt} + 1\},$$

where

$$\phi\left(\hat{\theta}_{ipt+1}; \hat{\theta}_{ipt}, \hat{\sigma}_{ipt}^2 - \hat{\sigma}_{ip,t+1}^2\right) = \frac{1}{\sqrt{2\pi (\hat{\sigma}_{ipt}^2 - \hat{\sigma}_{ip,t+1}^2)}} \exp\left\{-\frac{(\hat{\theta}_{ipt+1} - \hat{\theta}_{ipt})^2}{2 (\hat{\sigma}_{ipt}^2 - \hat{\sigma}_{ip,t+1}^2)}\right\}.$$

The variance term $\hat{\sigma}_{ipt}^2 - \hat{\sigma}_{ip,t+1}^2$ corresponds to the reduction in posterior uncertainty between periods t and $t+1$.

¹⁸The posterior mean is normally distributed because it is a linear function of Gaussian signals and a Gaussian prior, implying that belief updating preserves normality. The variance term follows from the law of total variance:

$$\begin{aligned} \text{Var}(\log \theta_{ip} \mid \hat{\theta}_{ipt}, \tau_{ipt}, a_{it}, \varphi_i) &= \mathbb{E}\left[\text{Var}(\log \theta_{ip} \mid \hat{\theta}_{ipt+1}, \tau_{ipt+1}, a_{it+1}, \varphi_i) \mid \hat{\theta}_{ipt}, \tau_{ipt}, a_{it}, \varphi_i\right] \\ &\quad + \text{Var}\left(\mathbb{E}[\log \theta_{ip} \mid \hat{\theta}_{ipt+1}, \tau_{ipt+1}, a_{it+1}, \varphi_i] \mid \hat{\theta}_{ipt}, \tau_{ipt}, a_{it}, \varphi_i\right), \end{aligned}$$

where $\text{Var}(\log \theta_{ip} \mid \hat{\theta}_{ipt}, \tau_{ipt}, a_{it}, \varphi_i) = \hat{\sigma}_{ipt}^2$, $\text{Var}(\log \theta_{ip} \mid \hat{\theta}_{ipt+1}, \tau_{ipt+1}, a_{it+1}, \varphi_i) = \hat{\sigma}_{ipt+1}^2$, and $\mathbb{E}[\log \theta_{ip} \mid \hat{\theta}_{ipt+1}, \tau_{ipt+1}, a_{it+1}, \varphi_i] = \hat{\theta}_{ipt+1}$.

2.3.6 Firm Value Function

Let $\Theta_{it} \equiv \left\{ \left(\hat{\theta}_{ip_k t}, \tau_{ip_k t} \right) \right\}_{k=1}^{n_i}$ denote firm i 's portfolio of posterior beliefs about product-specific expertise, where k indexes product lines and p_k denotes product p assigned to product line k . The firm's exit probability in the next period equals the probability that all posterior beliefs $\left\{ \mathcal{B}_{it+1}(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}) \right\}_{k=1}^{n_i}$ fall below the exit cutoff and all searches fail:

$$(1 - x)^{n_i} \prod_{k=1}^{n_i} \Pr \left(\mathcal{B}_{it+1}(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}) < \mathcal{B}_{it+1}^d(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}) \mid \mathcal{B}_{it}(\hat{\theta}_{ip_k t}, \tau_{ip_k t}) \right).$$

For vacant lines, this conditional probability degenerates to one, as a failed search leaves the line vacant.

Under Bayesian learning, these conditional probabilities do not reduce to unconditional probabilities. Consequently, product-line values are interdependent: posterior beliefs about product-specific expertise in other lines affect the exit probability and, hence, the value of product line k . We remove this dependence by assuming that the value of exit equals the value of dropping the product and continuing to search.¹⁹ Under this assumption, the following lemma establishes that product-line values are independent across lines:

Lemma 2 (Independence of Product Line Values). *Suppose that the value of a product line upon firm exit equals the value of dropping the product while continuing to search, i.e., $V^\emptyset(a_{it}, \psi_{it}, \varphi_i) = \beta V^{ss}(\mathbf{0}; a_{it+1}, \psi_{it+1}, \varphi_i)$. Then the value of product line k depends on its own state, $(\hat{\theta}_{ip_k t}, \tau_{ip_k t}; a_{it}, \psi_{it}, \varphi_i)$, and is independent of match qualities in other product lines, $V(\hat{\theta}_{ip_k t}, \tau_{ip_k t}; a_{it}, \psi_{it}, \varphi_i)$.*

Proof. See Appendix A.1. □

As a result, the firm value is the sum of the values of its product lines:

¹⁹This assumption also ensures that firms optimally exit when all products they produce fall below the drop threshold.

Proposition 1. *The firm-level value function is additively separable across product lines. In particular,*

$$V(\Theta_{it}; a_{it}, \psi_{it}, \varphi_i) = \sum_{k=1}^{n_i} V(\hat{\theta}_{ip_k t}, \tau_{ip_k t}; a_{it}, \psi_{it}, \varphi_i).$$

Proof. Follows from Lemma 2. □

2.4 Potential Startups

Each period, a large pool of potential startups makes entry decisions concurrently with incumbents' search decisions. Potential startups draw firm type φ_i from a stationary distribution $H(\varphi)$. Prior to entry, firms do not observe their realization of φ_i and make entry decisions based on the expected value of entry. Entry is undirected, and each potential startup enters only one product market, with the outcome realized at the end of the period. Upon entry, φ_i is revealed. Then, free entry implies that the expected value of entry equals the entry cost for all startups:

$$\mathbb{E}_\varphi [V^e(\varphi)] = \mathcal{F}^e. \quad (2.15)$$

Since

$$V_t^e(\varphi) = \beta \left[\pi^*(\hat{\theta}_{ip_k}^0, \tau_{ip_k}^0, 0, \psi_t, \varphi) + \mathbb{E}_{(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}) | (\hat{\theta}_{ip_k}^0, \tau_{ip_k}^0)} V_{it+1}(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}, 1, \psi_{t+1}, \varphi) \right],$$

it follows that

$$\mathbb{E}_\varphi \left[\pi^*(\hat{\theta}_{ip_k}^0, \tau_{ip_k}^0, 0, \psi_t, \varphi) + \mathbb{E}_{(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}) | (\hat{\theta}_{ip_k}^0, \tau_{ip_k}^0)} V_{it+1}(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}, 1, \psi_{t+1}, \varphi) \right] = \frac{\mathcal{F}^e}{\beta}.$$

Furthermore, upon entry, firms with sufficiently low realizations of φ_i exit immediately if their continuation value is negative. Thus, there exists an exit cutoff $\underline{\varphi}^e$ such that

$$\pi^*(\hat{\theta}_{ip_k}^0, \tau_{ip_k}^0, 0, \psi_t, \underline{\varphi}^e) + \mathbb{E}_{(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}) | (\hat{\theta}_{ip_k}^0, \tau_{ip_k}^0)} V_{it+1}(\hat{\theta}_{ip_k t+1}, \tau_{ip_k t+1}, 1, \psi_{t+1}, \underline{\varphi}^e) = 0.$$

The free entry condition pins down the equilibrium mass of startups, $\mathcal{M}_t^e$, in each period, given aggregate prices and wages.

After entry and initial production, age-one startups conduct multiple searches up to their capacity $n(\varphi)$. Since they have experience in only one product, they search over a set of size $N - 1$, with assignments coordinated without replacement as before.

2.5 Equilibrium

2.5.1 Labor Market Clearing

From equation 2.9, the optimal labor demand of firm i for product p at $t + 1$ is

$$\ell_{ipt+1}^*(\hat{\theta}_{ipt}, \tau_{ipt}, a_{it}, \varphi_i) = \left(\frac{\eta - 1}{\eta} \right)^\eta P_{pt+1}^\eta C_{pt+1} \left[\mathcal{B}(\hat{\theta}_{ipt}, \tau_{ipt}, a_{it}, \varphi_i) \right]^\eta z(\varphi_i)^{\eta-1} w_{t+1}^{-\eta}.$$

Thus, total labor demand for product p is

$$L_{pt+1}^* = \mathcal{M}_{pt} \left(\frac{\eta - 1}{\eta} \right)^\eta P_{pt+1}^\eta C_{pt+1} w_{t+1}^{-\eta} \int_{\mathcal{X}} [\mathcal{B}(\xi)]^\eta z(\varphi)^{\eta-1} \tilde{\mu}_{pt}(\xi) d\xi, \quad (2.16)$$

where $\mathcal{M}_{pt}$ is the mass of firms operating in product market p at the end of period t (equivalently, at the beginning of $t + 1$), and $\tilde{\mu}_{pt}$ is the corresponding probability density over states $\xi \equiv (\hat{\theta}, \tau, a, \varphi)$, and $\mathcal{X} \equiv \mathbb{R} \times \mathbb{N} \times \mathbb{N} \times \Phi$, where $\Phi \subset \mathbb{R}$ (is compact and) indexes firm types, is the product-line state space. The labor market clearing condition is

$$L = \sum_{p=1}^N L_{pt+1}^*. \quad (2.17)$$

2.5.2 Product-Line Distribution and Firm Distribution

Let $\mu_{pt}(A) \equiv \mathcal{M}_{pt} \tilde{\mu}_{pt}(A)$ denote the total mass of firms (equivalently, product lines) in market p with states lying in $A \subseteq \mathcal{X}$ measured at the end of period t (equivalently, at the beginning of $t + 1$) after the realization of θ_{ipt} and search outcomes. Then the law of motion at a point $(\hat{\theta}', \tau', a', \varphi)$ with $\tau' > 0$, which shows incumbent lines in market p

that remain in market p , is

$$\begin{aligned}\mu_{p,t+1}(\hat{\theta}', \tau', a', \varphi) &= (1 - \delta) \int_{\mathbb{R}} \left[\mathbf{1}_{\mathcal{R}_{t+1}^{ns}}(\hat{\theta}', \tau', a', \varphi) + (1 - x) \mathbf{1}_{\mathcal{R}_{t+1}^{ss}}(\hat{\theta}', \tau', a', \varphi) \right] \\ &\quad \times g(\hat{\theta}', \tau', a', \varphi \mid \hat{\theta}, \tau' - 1, a' - 1, \varphi) \mu_{pt}(\hat{\theta}, \tau' - 1, a' - 1, \varphi) d\hat{\theta},\end{aligned}\tag{2.18}$$

For $\tau' = 0$, $(\hat{\theta}_{ipt+1}, \tau_{ipt+1}, a_{ipt+1}, \varphi_i) = (\hat{\theta}^0, 0, a', \varphi)$ when incumbent firm i successfully switches its product at the end of $t + 1$ and land in product market p , or when potential startups enter market p :

$$\begin{aligned}\mu_{p,t+1}(\hat{\theta}^0, 0, a', \varphi) &= (1 - \delta) \sum_{q \neq p} \int_{\mathbb{R}} \sum_{\tilde{\tau}'} \frac{x}{N - 1} \mathbf{1}_{\mathcal{R}_{t+1}^{ss} \cup \mathcal{R}_{t+1}^d}(\hat{\theta}', \tilde{\tau}', a', \varphi) \\ &\quad \times g(\hat{\theta}', \tilde{\tau}', a', \varphi \mid \hat{\theta}, \tilde{\tau}' - 1, a' - 1, \varphi) \mu_{qt}(\hat{\theta}, \tilde{\tau}' - 1, a' - 1, \varphi) d\hat{\theta} \\ &\quad + (1 - \delta) \sum_{q \neq p} \frac{x}{N - 1} \mu_{qt}(\emptyset, \emptyset, a' - 1, \varphi) \\ &\quad + \frac{\mathcal{M}_{t+1}^e}{N} h(\varphi) \mathbf{1}\{a' = 0\}.\end{aligned}\tag{2.19}$$

$\frac{1}{N-1}$ is the adjusted probability that a product line in q lands on p .²⁰ h is the density associated with H , and $\mu_{qt}(\emptyset, \emptyset, a' - 1, \varphi)$ denotes the mass of vacant product lines associated with product q belonging to firms of age $a' - 1$ and type φ . The first term captures incumbent product lines that, after updating their beliefs at $t + 1$, fall in the switching region and successfully transition into market p . The second term captures vacant product lines that successfully search into market p . The last term captures startup entry into market p at $t + 1$. Since entry is undirected, each product market receives $\mathcal{M}_{t+1}^e/N$ mass of startups.

²⁰For firms with $a' \geq 2$ and φ , the feasible search set has size $N - n(\varphi)$. So a product line currently assigned to q lands on p with probability $\frac{1}{N - n(\varphi)}$. However, if the firm already operates in p through another line, this probability is zero. For a firm with scope $n(\varphi)$ that operates in q , the probability that it also operates in $p \neq q$ is $\frac{n(\varphi) - 1}{N - 1}$ under random assignment without replacement. Taking this into account, the adjusted probability that a product line in q lands on p is $\frac{1}{N - n(\varphi)} \times \frac{N - n(\varphi)}{N - 1} = \frac{1}{N - 1}$ by a law of large numbers with a continuum of firms. For firms with $a' = 1$ that operate in q , the feasible search set has size $N - 1$, and the firm cannot already operate in p . Hence, the landing probability is also $\frac{1}{N - 1}$.

Lastly, the mass of vacant product lines associated with product market p evolves according to:

$$\begin{aligned} \mu_{pt+1}(\emptyset, \emptyset, a', \varphi) = & (1 - \delta)(1 - p^e(a', \varphi)) \sum_{q \neq p} \int_{\mathbb{R}} \sum_{\tilde{\tau}'} \frac{1 - x}{N - 1} \mathbf{1}_{\mathcal{R}_{t+1}^d}(\hat{\theta}', \tilde{\tau}', a', \varphi) \\ & \times g(\hat{\theta}', \tilde{\tau}', a', \varphi \mid \hat{\theta}, \tilde{\tau}' - 1, a' - 1, \varphi) \mu_{qt}(\hat{\theta}, \tilde{\tau}' - 1, a' - 1, \varphi) d\hat{\theta} \\ & + (1 - \delta)(1 - p^e(a', \varphi)) \sum_{q \neq p} \frac{1 - x}{N - 1} \mu_{qt}(\emptyset, \emptyset, a' - 1, \varphi), \end{aligned} \quad (2.20)$$

where $p^e(a', \varphi)$ denotes the approximate probability that a firm of age a' and type φ has all non-focal product lines vacant and that all associated searches fail.²¹ Thus, the term $1 - p^e(a', \varphi)$ captures the probability that the firm does not exit endogenously, so that the vacant line remains in the state space.

The total mass of product lines (equivalently, firms) in market p is obtained by integrating the density (2.18)-(2.20) over all states. Since the transition density integrates to one, belief updating affects the distribution over states but not the aggregate mass.

Let the firm-level state be $\xi_{it}^f = (\Theta_{it}, a_{it}, \varphi_i)$, where $\Theta_{it} \equiv \{(\hat{\theta}_{ip_k t}, \tau_{ip_k t})\}_{k=1}^{n(\varphi_i)}$. Let $\Gamma_t(d\Theta, da, d\varphi)$ denote the firm-level cross-sectional distribution. The product-market-level measure μ_{pt} is the line-weighted projection of Γ_t . Thus the product-market distribution and the firm distribution are reconciled by a projection map, and scope heterogeneity enters only through the number of product lines contributed by each firm.

2.5.3 Symmetric Stationary Recursive Competitive Equilibrium

Definition 1. A symmetric stationary recursive competitive equilibrium consists of value functions and policy rules for firms, stationary prices w , $\{P_p\}_{p=1}^N$, and a stationary distribution of product lines $\mu = \{\mu_p\}_{p=1}^N$ such that: (i) given prices and μ , firms optimally choose continuation, search, and quantities for each product line, taking prices as given;

²¹While this event depends on the joint distribution of product-line states within the firm, tracking it explicitly would require a high-dimensional firm-level state space. To maintain tractability, we approximate the joint probability using the cross-sectional distribution of product-line states, $\tilde{\mu}_{pt}$. We assess the accuracy of this approximation in the simulation stage.

(ii) markets clear; (iii) the distribution μ is invariant under the law of motion induced by firms' policies, Bayesian updating, entry, and exit; and (iv) all product markets are identical.

2.6 Model Implications

2.6.1 Output Wedge under Imperfect Learning

Building on the biased learning result in Lemma 1, we define a wedge between perceived and true product-specific expertise to characterize how beliefs affect production decisions. This wedge captures how firms' posterior beliefs enter output through $\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1})$, allowing firm-level output to be expressed as a scaled version of the level implied by θ_{ip} .

Because firms with lower information-processing ability—when they are young—place greater weight on the prior and face noisier signals, they systematically underestimate high- θ_{ip} products and overestimate low- θ_{ip} products. As a result, these firms produce relatively less of high- θ_{ip} products and relatively more of low- θ_{ip} products compared to firms with higher ψ . Since households place greater weight on high- θ_{ip} products, these wedges lower welfare. The following proposition formalizes this relationship.

Proposition 2 (Learning Frictions and Output Wedges). *Define the wedge between perceived and true product-specific fundamentals as*

$$1 - \omega_{ipt} \equiv \frac{\left[\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) \right]^\eta}{\theta_{ip}}. \quad (2.21)$$

Then firm-level output can be written as

$$y_{ipt}^* = (1 - \omega_{ipt})\theta_{ip}z(\varphi_i)^\eta \left(\frac{\eta - 1}{\eta} \right)^\eta P_{pt}^\eta C_{pt} w_t^{-\eta}.$$

Firms with lower information-processing ability ψ_{it} exhibit systematic variation in ω_{ipt} : for a wider range of high- θ_{ip} products, $\omega_{ipt} > 0$, while for low- θ_{ip} products, $\omega_{ipt} < 0$. As a

result, these firms produce relatively less of high- θ_{ip} products and relatively more of low- θ_{ip} products compared to firms with higher ψ_{it} .

Proof. See Appendix A.1. □

2.6.2 Product Switching under Imperfect Learning

In addition to its effect on output, biased learning also influences firms' product-switching decisions by affecting the two switching thresholds. To illustrate this mechanism more transparently, and for expositional simplicity, we consider a two-period environment in which firms operate a single product and enter period $t=1$ with given posterior beliefs.

In period 1, after the realization of θ_{ip1} , firms update their posterior beliefs to $(\hat{\theta}_{ip1}, 1)$. They then decide whether to search, and conditional on the search outcome, choose quantities for the product operated in the next period. Period-2 profits are determined by the static profit maximization problem and therefore coincide with (2.13). Using this setup, we characterize the firm's value function in period at the search decision stage (after the realization of θ_{ip1}) as follows:

$$V_{i1}(\hat{\theta}_{ip1}, 1) = \max \left\{ V_{i1}^{ns}(\hat{\theta}_{ip1}, 1), V_{i1}^{ss}(\hat{\theta}_{ip1}, 1) \right\},$$

where

$$\begin{aligned} V_{i1}^{ns}(\hat{\theta}_{ip1}, 1) &= \beta \mathbb{E}_{\theta_{ip2}} \pi_{i2}(\hat{\theta}_{ip1}, 1) \\ V_{i1}^{ss}(\hat{\theta}_{ip1}, 1) &= -\mathcal{F}^{ss} + \beta \left[x \mathbb{E}_{\theta_{ip2}} \pi_{i2}(\hat{\theta}^0, 0) + (1-x) \mathbb{E}_{\theta_{ip2}} \pi_{i2}(\hat{\theta}_{ip1}, 1) \right]. \end{aligned}$$

Under the assumption that $\mathcal{B}_{i1}(\hat{\theta}^0, 0) > \mathcal{B}_{i1}^d$, we obtain the following proposition, which characterizes product-switching decisions as a function of firms' learning ability ψ_{i1} and productivity $z(\varphi_i)$.

Proposition 3 (Biased Learning and Product Switching). *Firms with lower information-processing ability ψ_i are more prone to operating in misaligned product lines due to biased*

learning. When the true θ_{ip} is high, they are more likely to replace or drop products that should be retained; whereas when θ_{ip} is low, they are more likely to retain products that should be dropped. Furthermore, this misalignment becomes more pronounced among firms with higher productivity $z(\varphi_i)$, i.e., the likelihood that firms operate misaligned product lines for a given ψ_i increases in $z(\varphi_i)$.

Proof. See Appendix [A.1](#). □

Since information-processing ability is closely linked to firm age, the model generates lifecycle patterns in product dynamics: when firms are young, they are less likely to retain products aligned with their expertise and more likely to retain misaligned products. These patterns suggest that firms make noisier product-portfolio decisions when they are young, due to limited information and lower information-processing ability.

3 Data and Measurement

In this section, we empirically test the model’s predictions. Our main data source is the quinquennial Census of Manufacturers (CMF, henceforth) linked to the Longitudinal Business Database (LBD, henceforth) hosted by the U.S. Census Bureau. Our sample covers the following eight quinquennial census years: 1977, 1982, 1987, 1992, 1997, 2002, 2007, and 2012.

The CMF provides comprehensive information on all U.S. manufacturing establishments with one or more paid employees. The data collects establishment-level characteristics such as employment, payroll, worker hours, payroll supplements, cost of materials, selected operating expenses, value added, capital expenditures, inventories, energy consumption, and industry codes. It also provides the list of input materials and expenses used by each establishment. Furthermore, it contains product codes—either five-digit SIC or seven-digit NAICS codes—and the value of shipments for products manufactured by each establishment.²² More details can be found from [Bernard et al.](#)

²²For example, in the Nonferrous Wiredrawing and Insulating industry (SIC 3357), there are thirteen

(2010), [Kehrig et al. \(2011\)](#) and [Kehrig and Vincent \(2021\)](#).

The LBD tracks the universe of U.S. private non-farm business establishments and firms with at least one paid employee. It covers all sectors and geographic areas of the economy annually from 1976 onward. Establishments owned by a parent firm are grouped under a common firm identifier, enabling us to aggregate establishment-level activities to the firm level. The LBD contains basic information such as employment, payroll, industry codes, and location. These variables enable us to link to the CMF establishments and identify parent firms to construct firm-level variables such as firm size, age, productivity, employment growth rates, and entry/exit. We refer to [Jarmin and Miranda \(2002\)](#), and [Chow et al. \(2021\)](#) for further details.

Industry-level variables are adopted from the NBER CES Manufacturing Industry Database, assembled by [Becker et al. \(2013\)](#). An industry is defined by the four-digit SIC (or the six-digit NAICS after 1997) code. The NBER-CES Manufacturing Industry Database contains annual industrial data for the U.S. manufacturing sector from 1958 through 2018, sourced from the U.S. Census Bureau, the Bureau of Economic Analysis, and the Bureau of Labor Statistics. The data covers industry-level information on price deflators, payroll, employment size, number of workers, total value of shipments, value added, various input costs and expenditures, and productivity (TFP estimates).

3.1 Product Definition and Characteristics

We define a product by a five-digit SIC code for the entire census years.²³ Note that in each Census year, there are two versions of product codes: one based on the previous Census year’s classification (“as collected”) and another based on the current Census year’s classification. For 1977–1997, the “current” classification is based on the 1987 SIC system. Starting from 2002, product codes based on the NAICS classification are

products, including Aluminum Wire (33571), Copper Wire (33572), Telephone Wire (3357B), and Fiber Optic Cable (33579).

²³This is based on product trailer information provided by the Census for the U.S. manufacturing firms in the CMF. We extract the information at the SIC5 or NAICS7 level for the pre-1997 and post-1997 Census years, respectively.

used. Importantly, in the 1997 CMF file, both the 1997 NAICS-based product code and the 1987 SIC-based product code appears, allowing us to construct a time-consistent product classification based on the 1987 SIC.

The bridge construction from the 1987 SIC to any version of NAICS follows a rolling approach. Starting with the 1987 SIC-to-1997 NAICS concordance, we then link the 1997 NAICS to the 2002 NAICS using the 2002 Census bridge file, yielding a concordance between the 1997 SIC and 2002 NAICS. We repeat this process for the 2007 and 2012 Censuses, ultimately generating a 1987 SIC to 2002, 2007, and 2012 NAICS concordance. To maximize within-firm consistency across years from 2002 to 2012, we first match each firm-product pair in a given year to its corresponding pair in the previous year. For unmatched cases, we use the bridge file constructed from the previous year to map the NAICS product code to its corresponding 1987 SIC product code. If a NAICS product code maps to multiple 1987 SIC codes, we select the one most frequently matched to that NAICS code.

The CMF reports product-level sales (product value of shipment) and physical quantities shipped. Given the definition of a product, we limit our analyses to observations that have a positive, nonzero value of shipments. Following literature, we drop “administrative record,” which are very small establishments that are not required to report the product-level value of shipments. We also exclude observations outside the “tabbed sample,” which is the sample used by the Census Bureau for official tabulations. We use the product value of shipment at the product-establishment level and aggregate it up to the product-firm level using LBD firm identifiers. We normalize the product value of shipments using the NBER-CES industry-level price indices (in 1997 U.S. dollars). For each product-firm pair, we measure tenure as the number of years the firm manufactures the product and appears in the record. To improve accuracy, we supplement our data with product-firm information from the Annual Survey of Manufacturers (ASM).

Lastly, for each firm in year t , we flag “added products” as those that were not present in the firm’s product portfolio before year t but appear by $t + 5$. In a similar fashion, we flag “dropped products” within each firm in year t as those that are present in the

firm’s product portfolio up to year t but no longer appear from $t + 5$. This enables us to identify added and dropped products from 1982 to 2007 in our sample.²⁴ For each added and dropped product, we indicate whether they are initial additions or re-entries, and whether drops are temporary or permanent. Our baseline analysis focuses on initial additions and permanent drops.

3.2 Firm Characteristics and Identifiers

We use firm age variables constructed by the Census using the method in [Haltiwanger et al. \(2013\)](#). Specifically, firm age is defined as the age of the oldest establishment owned by the firm when it is first observed in the data. We indicate young firms as those younger than or equal to age five. Note that because the LBD begins in 1976, all firms are assigned age zero in that year. To reduce misclassification of young firms, we use the 1972 CMF to adjust firm ages accordingly.²⁵ Firm size is measured using either total employment or total value of shipments.

One limitation of the LBD is the lack of longitudinally consistent firm identifiers. Although the redesigned LBD has a new firm identifier that links firms across time by correcting previous firm identifiers that are recycled in the old LBD, as described in [Chow et al. \(2021\)](#), it still has limitations.²⁶ However, longitudinal consistency of firm identifiers is necessary for our analysis to track firms’ history of product portfolios. Therefore, we construct alternative versions of longitudinal firm identifiers from the LBD following [Dent et al. \(2018\)](#) and use them to assess the robustness of our analysis results. Henceforth, we use the term “firm identifier” to refer to the firm identifiers constructed by the Census Bureau (lbdfid). All nominal values are converted to 1997 U.S. dollars using industry-level deflators for the relevant variables.

²⁴We use the 1977 data as a buffer to more accurately flag product additions and drops and compute product-firm tenure.

²⁵We use Finished Goods Inventory Beginning (FIB) from the 1972 CMF and add 4 to the firm age of LBD firms that own 1972 CMF establishments if FIB equals zero, and 5 years if FIB is greater than zero.

²⁶The new firm identifiers haven’t still resolved firm reorganization issue. See more discussion in [Chow et al. \(2021\)](#).

Table 1: Share of Product Switching Firms

variables	Including entry and exit			Continuers only		
	All	Age 0-5	Age 6 +	All	Age 0-5	Age 6 +
	Product adding only					
Share of firms	0.094	0.100	0.093	0.128	0.130	0.128
Output share	0.058	0.098	0.057	0.067	0.139	0.066
	Product dropping only					
Share of firms	0.398	0.329	0.407	0.182	0.143	0.187
Output share	0.282	0.404	0.280	0.169	0.146	0.169
	Product churning					
Share of firms	0.248	0.310	0.239	0.336	0.394	0.327
Output share	0.582	0.284	0.588	0.674	0.409	0.678

Notes: Firms in the manufacturing sector from 1982 to 2007.

4 Empirical Models and Main Results

In this section, we investigate firms' product-switching decisions and examine how the decisions differ for young firms. The design of our empirical strategy is discussed in the following subsections.

4.1 Prevalence of Product Switching Firms

Table 1 reports the share of product-switching firms and their output shares by age group, with and without entry and exit. Since firm entry and exit can overstate firm-level product additions and drops, the last three columns report shares for continuing firms only. Focusing on continuing firms, we find that 65% of firms in the pooled sample switch products at least once every five years, and they account for 91% of total output. Thus, product switching is a prevalent behavior, and firms that engage in it account for a substantial share of economic activity, as emphasized by [Bernard et al. \(2010\)](#).

Table 1 also shows that product switching often involves simultaneous add and drop of products (churning). Approximately 52% of product-switching firms (34% out of 65%) both add and drop products, while the remainder either add or drop, but not both. The life-cycle pattern of product switching exhibits distinct dynamics. In particular,

Table 2: Share of Multi-Product Firms

Classification	Share of firms			Output share		
	All	Age 0-5	Age 6 +	All	Age 0-5	Age 6 +
sic5	0.450	0.432	0.452	0.891	0.615	0.896
sic4	0.332	0.301	0.336	0.838	0.451	0.845
sic2	0.183	0.153	0.187	0.702	0.267	0.711

Notes: Firms in the manufacturing sector from 1982 to 2007.

product churning is more pervasive among young firms, while the share of drop-only cases is lower compared to their more mature counterparts.

Table 2 additionally reports the share of multi-product firms under alternative definitions of a product: the baseline five-digit SIC product code, the four-digit SIC industry code, and the two-digit SIC sector code. Consistent with existing findings, multi-product firms comprise slightly less than half of all firms (45%) but account for 89% of total output. These multi-product firms also operate across multiple industries and sectors. Multi-product firms are also prevalent among young firms, comprising 43% of them and generating 62% of their output.

In contrast, while single-product firms make up 55% of all firms, they account for only 11% of total output. Moreover, firms that produce a single product and never switch their product throughout the sample period represent 14.8% of firms but contribute just 2.8% of total output.

Lastly, Table 3 reports the means and standard deviations of the number of products firms produce, add, and drop, based on the full sample including single-product firms. On average, firms produce 2.2 products, add 1.2 products, and drop 0.9 products over five years.²⁷ Although young firms produce fewer products on average (1.6 vs. 2.3), they tend to add more products (1.6 vs. 1.2) while dropping fewer products (0.8 vs. 1.0) compared to mature firms.

²⁷Bernard et al. (2010) report that the average number of products firms produce rises to 3.5 when single-product firms are excluded.

Table 3: Number of Products Produced/Added/Dropped

	Including entry and exit			Continuers only		
	All	Age 0-5	Age 6 +	All	Age 0-5	Age 6 +
Number of products firms produce						
Mean	2.189	1.877	2.233	2.192	1.645	2.269
Stdev.	(2.896)	(1.509)	(3.041)	(3.153)	(1.377)	(3.321)
Number of products firms add						
Mean	1.27	1.831	1.189	1.244	1.605	1.194
Stdev.	(1.831)	(1.518)	(1.858)	(1.880)	(1.374)	(1.935)
Number of products firms drop						
Mean	1.273	1.218	1.281	0.941	0.790	0.963
Stdev.	(1.981)	(1.482)	(2.043)	(1.841)	(1.101)	(1.922)

Notes: Firms in the manufacturing sector from 1982 to 2007.

4.2 Product-Specific Expertise Estimation

We estimate product-specific expertise of firms, which is interchangeably labeled “firm-product match quality,” and examine the relationship between firms’ product-switching decisions and resource reallocation. In the CMF, the only performance measure available at the product-firm level is the product value of shipments (PV).²⁸ However, using PV directly to assess whether a firm is relatively better at producing one product over another presents several issues. The level of PV may reflect not only firm-specific capabilities but also factors such as the duration of production (e.g., life-cycle pattern of product as in [Argente et al. \(2024\)](#) or customer capital accumulation) and transitory product-level shocks (e.g., market fluctuations).

To address these concerns, we estimate a time-invariant measure of product-firm performance by taking out factors that influence the level of PV other than the firm’s product-specific ability. We refer to this measure as firms’ product-specific expertise, or the match quality (fit) between a firm and a product, which captures the firm’s capability to produce a given product. We estimate the following baseline model, similar to

²⁸While quantity data are also available, they cover only a limited set of products.

Bonhomme et al. (2019):

$$y_{ipjt} = \theta_{ip} + X_{ipjt}\gamma_1 + X_{ijt}\gamma_2 + \delta_{pt} + \varepsilon_{ipjt}, \quad (4.22)$$

y_{ipjt} is the log real PV for firm i (in industry j) and product p in census year t . θ_{ip} , a firm-product fixed effect, is our main interest. This contains time-invariant characteristics associated with the firm-product match that determine the performance of the product within the firm and capture the match quality between them.

We include a rich set of controls and fixed effects to account for sources affecting the firm-product-specific value of shipments, unrelated to the firm-product-specific unobservable characteristics. For example, when firms enter new product markets, it typically takes time to build a customer base, which can affect shipment values independently of match quality. To control for this effect, we include product-firm-specific tenure as of year t and its square. Furthermore, we include a vector of firm-level time-varying characteristics (X_{ijt}), which includes firm size (log employment), firm age, total assets, number of products, number of establishments, cost of advertisements, cost of software, skill and capital intensity (log ratio of non-production workers to total employment and capital to total employment), productivity, and firm-level inputs (log of production workers, equipment capital stock, structural capital stock, material cost, and energy costs).

Lastly, δ_{pt} is a product-year fixed effect to control for any potential factors affecting the value of shipment for firm i 's product p , attributed to time-varying technology or demand structure that differs across each product market. Related to this, several product-specific properties (skill or capital intensity) can also affect the total value of shipment regardless of the firm-product's fundamental match quality. As robustness checks, we further explore alternative specifications to estimate the product-firm match quality. More discussion can be found in the Appendix.

Note that as we control for factors specific to product markets and firms each year, our measure for the product-firm match quality, θ_{ip} , can be compared across different

product-firm pairs across different years. We rescale the quality measure to make it normalized and range in $[0,1]$. Let q_{ip} denote this. In particular, we use the following formula to rescale $\hat{\theta}_{ip}$:

$$q_{ip} \equiv \frac{\hat{\theta}_{ip} - \underline{\theta}_{ip}}{\bar{\theta}_{ip} - \underline{\theta}_{ip}}, \quad (4.23)$$

where $\underline{\theta}_{ip}$ and $\bar{\theta}_{ip}$ are the minimum and maximum value of the quality estimates, respectively (e.g. $\hat{\theta}_{ip} \in [\underline{\theta}_{ip}, \bar{\theta}_{ip}]$). We estimate θ_{ip} using the CMF for the years 1977 to 2012, complemented with the ASM data from 1976 to 2016.

4.3 Product Dropping

In this subsection, we analyze the types of products that firms drop from their existing portfolios and examine how the pattern of dropping products vary between young and mature firms.

4.3.1 Match Quality of Dropped Products

We begin by examining the types of products that firms drop. Specifically, we test whether firms are more likely to drop products with lower performance—that is, products that have poorer firm-product match quality—relative to better-matched ones. This is based on the premise that identifying match quality of products is not straightforward, given the costs associated with searching for and learning about the quality of a product-firm match. This challenge may be more pronounced for younger firms, which lack enough records or information about themselves and products in the market.

To see this, we construct a dummy variable, $\mathcal{I}_{ipjt}^{drop}$, to indicate a product being dropped by the beginning of the next period. $\mathcal{I}_{ipjt}^{drop}$ is equal to one if firm i in industry j drops product p between period t and $t + 5$ (e.g., if product p appears in firm i 's portfolio in t but not $t + 5$). We regress it on the product-firm performance measures to see how

Table 4: Probability of Product Drop

	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$
$\log(PV_{pit})$	-0.052*** (0.002)			-0.052*** (0.002)	
Quality _{pi}		-1.074*** (0.021)			-1.081*** (0.022)
Young Firm _{it}			-0.075*** (0.003)	-0.141*** (0.007)	-0.368*** (0.022)
$\log(PV_{pit}) \times \text{Young}_{it}$				0.010*** (0.001)	
Quality _{pi} $\times$ Young _{it}					0.497*** (0.043)
Observations	1,622,000	1,622,000	1,622,000	1,622,000	1,622,000
Fixed effects	pt, i	pt, i	pt, i	pt, i	pt, i
Controls	Full	Full	Full	Full	Full

Notes: Estimates for product-year, firm fixed effects, and the constant are suppressed. Robust standard errors clustered at the five-digit SIC product level are reported in parentheses. Observations are unweighted, and observation counts are rounded due to the Census Bureau disclosure avoidance procedures. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

firms' product drop depends on the match quality between product p and firm i :

$$\mathcal{I}_{ipjt}^{drop} = \alpha + \beta Q_{ipt} + X_{ipjt}\gamma_1 + X_{ijt}\gamma_2 + \delta_i + \delta_{pt} + \varepsilon_{ipjt}. \quad (4.24)$$

Q_{ipt} is either the log of real PV or q_{ip} , the match quality value (4.23) associated with the pair of product p and firm i . X_{ipjt} is a vector of time-varying firm-product characteristics (log firm-product specific tenure and square, and log value of shipment if q_{ip} is used for Q_{ipt}). X_{ijt} is a vector of firm controls (the same set of controls used in q_{ip} estimation). δ_{pt} is a product-year fixed effect, and δ_i is a firm fixed effect.

The first two columns of Table 4 present results from specifications that include the full set of control variables.²⁹ As shown in column 1 of Table 4, firms are, on average, more likely to drop products that generate lower revenue. Column 2 further shows that this pattern holds for our time-invariant firm-product performance measure: firms

²⁹The corresponding full model results are reported in Table A1 and A2 in Appendix B.1.

tend to drop products with lower match quality, relative to better-matched ones. These results suggest that, on average, firms are able to identify which products are better matches and drop those that are poorly matched. Dropping poorly matched products would free up the resources and potentially enable firms to reallocate these resources toward their better use. Before examining whether dropping poor-matched products helps firms climb the match-quality ladder and improve their overall performance, we test whether product-dropping behavior varies by firm age.

4.3.2 Product Dropping by Young Firms

As young firms have less experience in producing and selling products, they may struggle to accurately assess whether a product is a good match for them. This limited information and evaluation capacity of young firms could make them drop products less frequently than old firms, particularly those with poor match quality.

To investigate such heterogeneity in product-dropping behavior among young firms, we first estimate the following regression model:

$$\mathcal{I}_{ipjt}^{drop} = \alpha + \beta \mathcal{I}_{ijt}^{young} + X_{ipjt}\gamma_1 + X_{ijt}\gamma_2 + \delta_i + \delta_{pt} + \varepsilon_{ipjt}, \quad (4.25)$$

where $\mathcal{I}_{ijt}^{young}$ is a dummy variable equal to one if a firm is with age less than or equal to five in year t .

One thing to note is that single-product firms could choose not to drop their product even though they know that the product they produce is a poor match for them. This is because dropping the product would force them to shut down their business, as it is their only product. Because young firms are more likely single-product firms, including the number of products firms produce as a firm-level control is particularly critical in this regression model.³⁰ Financial frictions can also matter. The product-switching behavior of financially constrained firms may differ from that of unconstrained firms,

³⁰As shown in Table 2, young firms are approximately 4% less likely to be multi-product firms compared to mature firms (43% vs. 45%).

as the former face greater difficulty in financing new projects to add products. One way to mitigate this concern is to include a proxy for firms' ability to raise funds. The closest available variable in the CMF is total assets.³¹ We include these variables, as well as other firm controls as before. The third column of Table 4 presents the result. This shows that on average, young firms drop products less frequently than mature firms.³²

To further explore the match quality of products dropped by young firms, we interact the two firm-product performance measures in equation (4.24) with the young firm dummy as follows:

$$\mathcal{I}_{ipjt}^{drop} = \alpha + \beta_1 Q_{ipt} + \beta_2 \mathcal{I}_{ijt}^{young} + \beta_3 Q_{ipt} \times \mathcal{I}_{ijt}^{young} + X_{ipjt} \gamma_1 + X_{ijt} \gamma_2 + \delta_i + \delta_{pt} + \varepsilon_{ipjt}. \quad (4.26)$$

Columns 4 and 5 of Table 4 lay out the results.³³ In particular, the third row (the coefficient on the young firm dummy) indicates that with match quality controlled, young firms drop products even less frequently than their mature counterparts on average.

However, as shown in the fourth and fifth rows, young firms are more likely to drop products in which they perform well while dropping fewer products with poor match quality, relative to mature firms. In other words, young firms tend to drop products they should have retained while retaining products they should have dropped. Overall, the stickiness with poor match quality products is the dominant factor determining the average product-dropping behavior of young firms.

These results suggests that young firms' limited information for assessing product match quality may constrain their ability to keep better-matched products while dropping poorly matched ones at the early stage of their life-cycle. As a result, young firms may struggle to climb the product match quality ladder. Such a restriction can hinder the potential growth of young firms, especially in environments with greater noise and

³¹We are in the process of exploring another way to address financial frictions, such as industry-level differences in external financing arising from the relative importance of intangible versus physical capital.

³²Table A3 in Appendix B.1 shows the results for including and excluding firm-level controls.

³³Table A4 and A5 in Appendix B.1 show the results for the full model.

less accurate information for firms to learn, such as during recessions.³⁴

4.4 Firm-Level Evidence for Quality Ladder

Next, we examine whether and how firms climb the match-quality ladder.

4.4.1 Product Adding

We begin by examining whether the probability of adding a new product depends on the firm's prior product-dropping behavior. To see this, we define a product as being added to a firm's portfolio in year t if it is absent in the firm's portfolio in t but appears by the next census year $t + 5$. Let $\mathcal{I}_{ijt}^{add}$ denote a dummy variable equal to one if firm i in industry j adds at least one product between t and $t + 5$.

We estimate the following conditional probability of adding at least one product conditional on having at least one product dropped, with other factors controlled as before:

$$\mathcal{I}_{ijt}^{add} = \alpha + \beta \mathcal{I}_{ijt}^{drop} + X_{ijt} \gamma + \delta_i + \delta_{jt} + \varepsilon_{ijt}. \quad (4.27)$$

$\mathcal{I}_{ijt}^{drop}$, is a dummy variable equal to one if firm i dropped at least one of its products between years t and $t + 5$. δ_i is a firm fixed effect, and δ_{jt} is an industry-year fixed effect. The remaining firm-level controls are identical to those used previously.

As shown in column 1 of Table 5, the probability that a firm adds a new product is positively associated with their simultaneous dropping of at least one product.^{35,36} Columns 2 and 3 report results from a regression specification that extends equation (4.27) by interacting the product-dropping dummy with the young firm dummy. The estimates show that young firms, on average, are more likely to add products than their mature counterparts, and this is primarily driven by their higher propensity to add products when simultaneously dropping others. This indicates that firms simultaneously add

³⁴We also find consistent results over the business cycle, where such patterns of product switching for young firms are more pronounced in recessions. Results are available on request.

³⁵Table A6-A9 in the Appendix show the full results.

³⁶This result is consistent with Bernard et al. (2010).

Table 5: Product Add and Drop

	$Add_{t:t+5}$	$Add_{t:t+5}$	$Add_{t:t+5}$
$Drop_{t:t+5}$	0.096*** (0.012)		0.070*** (0.012)
Young Firm $_{it}$		0.148*** (0.005)	0 (0.005)
$Drop_{t:t+5} \times \text{Young}_{it}$			0.295*** (0.008)
Observations	682,000	682,000	682,000
Fixed effects	i, jt	i, jt	i, jt
Controls	Full	Full	Full

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Robust standard errors clustered at the firm's modal five-digit SIC product level are reported in parentheses. Observations are unweighted, and observation counts are rounded due to the Census Bureau disclosure avoidance procedures. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

and drop products, with young firms exhibiting a higher propensity to do so, which is a piece of evidence regarding the match-quality ladder.³⁷

4.4.2 Match Quality of Added Products: Quality Ladder Evidence

In the previous analyses, we found that the propensity to add a new product is positively associated with the probability of dropping products at the firm level. In this section, we further investigate whether the match quality of newly added products is associated with firms' product dropping behavior to assess whether firms climb up the match-quality ladder through product switching.

To do so, we define $\mathcal{I}_{ijt}^{add,high}$, a dummy variable equal to one if the average match quality of products newly added between t and $t + 5$ is higher than the average match quality of the products firms produce in t . Using this variable, we examine whether the probability of adding a product with higher match quality than the existing ones is greater when firms simultaneously drop products. If firms are indeed climbing the match-quality ladder, we would expect this relationship to be positive. We estimate

³⁷We find similar results when using the number of products added as the dependent variable. The results are reported in Table A10 in the Appendix.

Table 6: Relative Match Quality of Added Products

	$Add_{t:t+5}^{high}$	$Add_{t:t+5}^{high}$	$Add_{t:t+5}^{high}$
$Drop_{t:t+5}$	0.028** (0.011)		0.049*** (0.010)
Young Firm _{it}		-0.112*** (0.017)	0.044* (0.024)
$Drop_{t:t+5} \times \text{Young}_{it}$			-0.228*** (0.024)
Observations	66,000	66,000	66,000
Fixed effects	i, jt	i, jt	i, jt
Controls	Full	Full	Full

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Robust standard errors clustered at the firm's modal five-digit SIC product level are reported in parentheses. Observations are unweighted, and observation counts are rounded due to the Census Bureau disclosure avoidance procedures. * p<0.1, ** p<0.05, *** p<0.01.

the following regression model using the sample of firms that add at least one product between t and $t + 5$:

$$\mathcal{I}_{ijt}^{add,high} = \alpha + \beta \mathcal{I}_{ijt}^{drop} + X_{ijt}\gamma + \delta_i + \delta_{jt} + \varepsilon_{ijt}. \quad (4.28)$$

The fixed effects and firm-level controls are identical to those used previously.

As shown in the first column of Table 6, the probability of adding products with match quality higher than that of existing products becomes higher if at least one product is dropped within the same period.³⁸ This result confirms that, on average, not only the likelihood of adding a product but also the match quality of the added products is positively associated with firms' product-dropping behavior. This is consistent with the match-quality ladder hypothesis.

We then examine whether the relationship between the match quality of added products and product drop differs for young firms by interacting the drop dummy with the young firm dummy. Columns 2 and 3 of Table 6 show that young firms are less likely to add higher match quality products relative to their mature counterparts, and this dif-

³⁸See Table A11-A14 in the Appendix for the full results.

ferential is primarily driven by instances in which they simultaneously drop products. As shown in column 3 of Table 5, young firms tend to add products more frequently when they simultaneously drop others. Taken together with the previous findings, this suggests that young firms face greater difficulty in identifying and retaining the right products. While they actively experiment by adding new products, many of these additions do not align well with their underlying expertise.

However, when young firms add new products without simultaneously dropping any, they are more likely to add products with higher match quality. Table 1 shows that product adding accompanied by simultaneous dropping (churning) occurs three times more frequently than adding without dropping. Moreover, the average match quality of products added with simultaneous dropping is substantially higher than that of products added without dropping.³⁹ Therefore, despite the positive outcome in the case of adding without dropping, it is outweighed by the more prevalent and dominant negative case in which young firms churn products.

This provides additional evidence consistent with the match-quality ladder hypothesis. Combined with the previous findings, this suggests that firms learn about their expertise while operating, enabling them to reoptimize their product portfolios by adding products that are better matched to their capabilities while dropping the ones that are not. However, young firms appear to face challenges in this process: they are less likely to add products with higher match quality than their existing portfolio and more likely to drop products they should retain. In other words, young firms struggle to climb the match-quality ladder.

4.5 Proximity of Added Products to Existing Products

Up to this point, we have focused on the relationship between product dropping and adding, as well as the relative match quality of products. We now turn to examining the proximity between newly added products and a firm's existing product portfolio.

³⁹We are currently in the process of disclosing this result again. Thus, it is not included in the current draft.

Table 7: Number of Input Materials Used per Product by Firms

	Including entry and exit			Continuers only		
	all	age 0-5	age 6 +	all	age 0-5	age 6 +
Mean	20.30	19.45	20.43	21.33	21.50	21.31
Stdev.	(20.29)	(19.86)	(20.35)	(21.00)	(21.34)	(20.95)

Notes: Firms in the manufacturing sector from 1982 to 2007.

Table 8: Material-Based Closeness of Added Products

	$Mat_{t:t+5}^{add}$	$Mat_{t:t+5}^{add}$	$Mat_{t:t+5}^{add}$
$Drop_{t:t+5}$	0.098*** (0.004)		0.102*** (0.004)
Young Firm _{it}		-0.003 (0.004)	0.022*** (0.005)
$Drop_{t:t+5} \times \text{Young}_{it}$			-0.037*** (0.006)
Observations	198,000	198,000	198,000
Fixed effects	i, jt	i, jt	i, jt
Controls	Full	Full	Full

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Robust standard errors clustered at the firm's modal five-digit SIC product level are reported in parentheses. Observations are unweighted, and observation counts are rounded due to the Census Bureau disclosure avoidance procedures. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

The literature finds that firms tend to produce products close to each other in terms of various measures, such as industry classification or the composition of material inputs used (e.g., [Bernard et al., 2010](#); [Boehm et al., 2022](#)). Nonetheless, we also observe many instances in which firms expand into entirely new industries.

To examine whether the proximity of added products to a firm's existing product portfolio varies over the firm life-cycle, we use the share of input materials used to produce a newly added product that are also used for the firm's existing products as a proxy for product proximity.^{40,41} Let $MaterialCloseness_{ijt}^{add}$ denote the average of

⁴⁰The CMF provides establishment-level data on the list of material inputs used and associated expenditures. By focusing on single-product establishments, we identify the set of material inputs used in the production of each product.

⁴¹We are in the process of exploring an expenditure-weighted version of the material input-based

the proximity measure of products added by firm i between t and $t + 5$. The mean (standard deviation) of the material-based closeness of added products in the churning case is 0.708 (0.271). Table 7 shows the number of input materials used per product by firms across age groups. Focusing on the continuing firms, we find that young firms use slightly more variety of input materials to produce each product, and the variance is also larger compared to mature firms.

We then estimate the following regression model:

$$MaterialCloseness_{ijt}^{add} = \alpha + \beta \mathcal{I}_{ijt}^{drop} + X_{ijt}\gamma + \delta_i + \delta_{jt} + \varepsilon_{ijt}. \quad (4.29)$$

Column 1 of Table 8 shows that, on average, firms add products that are closer to their existing portfolio when they simultaneously drop products. However, this pattern differs across age groups. Column 3 presents regression results from an extended specification that interacts the drop dummy with the young firm dummy. The results indicate that, relative to mature firms, young firms tend to add products that are more distant from their existing product portfolio.⁴² Table A19 in the Appendix presents the results using industry-based proximity, and the findings are qualitatively similar.

4.6 Product Switching and Firm Performance

In this section, we further examine whether firms' product-adding or dropping behavior affects firm performance.

4.6.1 The Impact of Product Adding and Dropping

We begin by using the following regression model to estimate the relationship between the decision to add or drop a product and firm performance:

$$\Delta y_{ijt} = \alpha + \beta_1 \mathcal{I}_{ijt}^{drop} + \beta_2 \mathcal{I}_{ijt}^{add} + X_{ijt}\gamma + \delta_i + \delta_{jt} + \varepsilon_{ijt}, \quad (4.30)$$

proximity measure, following Boehm et al. (2022).

⁴²See Table A15-A18 in the Appendix for the full results.

Table 9: Product Add/Drop and Firm Performance

	ΔV_{add}	ΔEmp	ΔTFP	ΔLP
$Add_{t:t+5}$	0.830*** (0.019)	0.802*** (0.019)	0.735*** (0.019)	0.762*** (0.019)
$Drop_{t:t+5}$	-0.754*** (0.019)	-0.731*** (0.018)	-0.686*** (0.017)	-0.713*** (0.018)
Observations	682,000	682,000	648,000	648,000
Fixed effects	i, jt	i, jt	i, jt	i, jt
Controls	Full	Full	Full	Full

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Robust standard errors clustered at the firm's modal five-digit SIC product level are reported in parentheses. Observations are unweighted, and observation counts are rounded due to the Census Bureau disclosure avoidance procedures. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

where Δy_{ijt} is the log difference of i) real value added, ii) total employment, iii) total factor productivity, or iv) labor productivity between t and $t + 5$.⁴³

Table 9 presents the result, indicating that product dropping is negatively associated with subsequent growth across the four performance measures, while product adding is positively correlated with them.⁴⁴ When considering product switching—defined as the combination of adding and dropping products—it is positively associated with subsequent changes in all four performance measures.⁴⁵

4.6.2 Decomposition of Firm-Level Growth in Output and Match Quality

To understand the importance of within-firm reallocation of resources through product switching to firm growth further, we conduct an accounting decomposition of output and match quality growth at the firm-product level. This decomposition allows us to distinguish between the effects of changes in the share of output across products and output and match quality growth within each firm-product pair.

⁴³Value added is Total Value of Shipments + Final Inventory Investment + Work-in-Progress Goods - Resales - Material Inputs - Energy Expenditures. Labor productivity is the Real value of shipments/Total employment. We use the Census TFP beta 8 available in the CMF.

⁴⁴Table A20 in the Appendix reports the full set of results.

⁴⁵These results are consistent with previous findings by Bernard et al. (2010) and Argente et al. (2018).

Table 10: Firm-Level Output and Match Quality Growth Decomposition

	Output			Match quality		
	all	age 0-5	age 6 +	all	age 0-5	age 6 +
Total changes	0.112	0.662	0.104	0.017	0.130	0.016
Within share	0.543	0.444	0.545	0	0	0
Between share	0.008	0.002	0.009	0.318	0.126	0.320
Cross share	0.009	-0.005	0.010	0	0	0
Addition share	0.424	0.554	0.422	0.574	0.853	0.572
Drop share	0.015	0.005	0.015	0.107	0.022	0.108

Notes: Firms in the manufacturing sector from 1982 to 2007.

Define within-firm output share of product p at t for firm i as $\omega_{ipt} = \frac{PV_{ipt}}{\sum_{p \in \mathcal{P}_{it}} PV_{ipt}}$, where PV_{ipt} denotes the real product value of shipments of product p by firm i at t , and $\mathcal{P}_{it}$ is the set of products in firm i 's product portfolio at t . Let y_{ipt} denote either the log of real product value of shipments, $\log(PV_{ipt})$, or the firm-product match quality, q_{ip} . We compute the firm-level average output or match-quality as $y_{it} = \sum_{p \in \mathcal{P}_{it}} \omega_{ipt} y_{ipt}$.

We then decompose the growth in firm-level output or match quality as follows:

$$\begin{aligned}
 \Delta y_{it} = & \underbrace{\sum_{p \in \mathcal{C}_{it}} \omega_{pit-5} \Delta y_{ipt}}_{\text{within term}} + \underbrace{\sum_{p \in \mathcal{C}_{it}} (y_{pit-5} - y_{it-5}) \Delta \omega_{ipt}}_{\text{between term}} + \underbrace{\sum_{p \in \mathcal{C}_{it}} \Delta \omega_{ipt} \Delta y_{ipt}}_{\text{cross term}} \\
 & + \underbrace{\sum_{p \in \mathcal{N}_{it}} \omega_{ipt} (y_{ipt} - y_{it-5})}_{\text{addition term}} - \underbrace{\sum_{p \in \mathcal{X}_{it}} \omega_{pit-5} (y_{pit-5} - y_{it-5})}_{\text{drop term}}. \tag{4.31}
 \end{aligned}$$

$\Delta X_{.t} = X_{.t} - X_{.t-5}$ is the change in any variable $X_{.t}$ between $t-5$ and t , $\mathcal{C}_{it}$ is the set of continuing products that firm i produces in both $t-5$ and t , $\mathcal{N}_{it}$ is the set of products newly added by firm i in t , and $\mathcal{X}_{it}$ is the set of products dropped by firm i in t .

Table 10 presents the decomposition results for both output (real product value of shipments) and match quality. We normalize each term by the total change to quantify its contribution to overall growth. We refer to these normalized values as shares. The first three columns show that output growth is primarily driven by within-product

growth (within share) and new product addition (addition term). For young firms, however, new product addition plays a more prominent role than within-product growth. In fact, the combined contribution of product-switching components, addition and drop shares, is 28% higher for young firms than for mature firms ($0.554 + 0.005$ vs. $0.422 + 0.015$).

Taken together, these results provide suggestive evidence that firms foster growth by reallocating resources through product switching, effectively climbing the match-quality ladder. However, young firms struggle in this process, often devoting time and resources to discovering their areas of expertise by experimenting with a wider range of products.

5 Robustness Tests

As we show in the Appendix, our results remain robust to different sets of control variables. The results are also robust to using different firm identifiers and alternative specifications for estimating the match quality measure. For analyses of firms' product-dropping behavior, product drops due to firm exit could affect our results in various ways. To address this concern, we re-estimate our models after excluding all exiting firms. The results remain robust.⁴⁶

Financial frictions may provide an alternative explanation for some of our results. To mitigate this concern and focus on the learning channel, we control for total assets in the current analysis and are exploring additional approaches to test and account for this mechanism. One of the alternative ways is to use industry-level variation in external funding, such as differences in physical versus intangible capital intensity.

⁴⁶The results using different firm identifiers, alternative match quality estimations, and restricting the sample to continuing firms only are available upon request.

6 Quantitative Analysis

TBA

6.1 Misallocation and Policy Implications (In Progress)

In this environment, misallocation arises because high-productivity firms operate in product markets in which they have relatively low expertise, which depresses demand for their products. Accelerating learning, increasing search intensity, and enabling firms to optimize their product portfolios enhance allocative efficiency.

Part of the role of learning operates as in Boyan Jovanovic’s framework: faster learning increases the speed of selection by accelerating the exit of low-performing firms. Our contribution is to introduce endogenous product portfolio optimization, which provides an additional margin of adjustment by reallocating firms toward product markets in which they have good expertise and face higher demand. The model allows us to quantify the relative contributions of selection and portfolio optimization to improvements in allocative efficiency.

Policies that expand access to data, improve managerial practices, or reduce search costs can facilitate these adjustments and thereby increase aggregate efficiency.

7 Concluding Remarks

In this paper, we develop a model of product switching over the firm lifecycle in which firms learn about product-specific expertise and adjust their product portfolios accordingly. Firms’ ability to process information improves over time, so imperfect learning early in the lifecycle generates systematic biases in beliefs. These biases create wedges in within-firm resource allocation across products and shape product-switching decisions. As a result, firms adjust their product portfolios more slowly toward products in which they can perform well when they are young, with implications for firm growth

and performance.

We take the model to a comprehensive administrative dataset that tracks U.S. manufacturing output at the firm–product level and document several key patterns. Firms tend to drop products with low product-specific expertise, but this pattern is weaker among young firms. In particular, young firms are more likely to retain low-expertise products and to drop high-expertise products. We also find that the likelihood of adding products with higher expertise than existing ones increases when firms simultaneously drop products, but this reoptimization margin is less pronounced among young firms. Taken together, these findings suggest that although young firms actively experiment by adding new products, they move more slowly toward products in which they perform well.

The model provides a unified interpretation of these patterns. Firms with lower information-processing ability—such as young firms—place greater weight on prior beliefs and rely on noisier signals, making them more likely to retain poorly performing products and drop better-performing ones. This mechanism links lifecycle patterns in product switching to differences in firm performance and growth.

These findings have broader implications for resource allocation. By shaping how firms adjust their product portfolios, learning frictions introduce a within-firm margin that affects the allocation of resources across product markets over time. This perspective suggests that policies aimed at improving firms’ ability to process information—such as investments in data infrastructure, managerial practices, or organizational capabilities—can accelerate learning and improve product-switching decisions. Moreover, if information-processing ability depends on spillovers across firms, these dynamics may generate externalities that further justify policy intervention. A more complete analysis of these implications is left for future work.

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Appendix

A Theory

A.1 Proofs

Proof of Lemma 1, Biased Learning

Proof. The first derivative of $\hat{\theta}_{ipt}$ with respect to ψ_{it} is given by

$$\frac{\partial \hat{\theta}_{ipt}}{\partial \psi_{it}} = \frac{(\hat{\sigma}_{ipt}^2)^2}{\sigma_\varepsilon^2} \left[\log s_{ipt} \left(\frac{1}{\sigma_0^2} + \frac{\tau}{\sigma_\varepsilon^2} + \frac{\sum_{\tilde{t}} \psi_{i\tilde{t}}}{\sigma_\varepsilon^2} \right) - \left(\frac{\theta_0}{\sigma_0^2} + \frac{\sum_{\tilde{t}} \log \theta_{ip\tilde{t}}}{\sigma_\varepsilon^2} + \frac{\sum_{\tilde{t}} \psi_{i\tilde{t}} \log s_{ip\tilde{t}}}{\sigma_\varepsilon^2} \right) \right].$$

Since $\log \theta_{ipt} = \log \theta_{ip} + \varepsilon_{ipt}$, where $\varepsilon_{ipt} \sim N(0, \sigma_\varepsilon^2)$, and $\log s_{ipt} = \log \theta_{ip} + v_{ipt}$, where $v_{ipt} \sim N(0, \frac{\sigma_\varepsilon^2}{\psi_{it}})$, taking expectations with respect to the noise terms yields

$$\mathbb{E}_{\varepsilon, v} \left(\frac{\partial \hat{\theta}_{ipt}}{\partial \psi_{it}} \right) = \frac{(\hat{\sigma}_{ipt}^2)^2}{\sigma_\varepsilon^2 \sigma_0^2} (\log \theta_{ip} - \theta_0).$$

Therefore, on average, when $\log \theta_{ip} > \theta_0$, a higher ψ_{it} raises the posterior mean, implying that firms with lower ψ_{it} underestimate $\hat{\theta}_{ipt}$. Conversely, when $\log \theta_{ip} < \theta_0$, a higher ψ_{it} lowers the posterior mean, so firms with lower ψ_{it} overestimate $\hat{\theta}_{ipt}$. Moreover,

$$\frac{\partial \hat{\sigma}_{ipt}^2}{\partial \psi_{it}} = -\frac{(\hat{\sigma}_{ipt}^2)^2}{\sigma_\varepsilon^2} < 0.$$

Therefore, firms with lower information-processing ability have a higher posterior variance. □

Proof of Lemma 2, Independence of Product Line Values

Proof. The value of product line k is determined by flow profits, the switching decision, and the firm's exit decision. Flow profits depend only on $(\hat{\theta}_{ipkt}, \hat{\sigma}_{ipkt}, \theta_{ipkt}, \varphi_i)$, and the

evolution of $(\hat{\theta}_{ipt}, \hat{\sigma}_{ipt}, \theta_{ipt})$ depends only on firm-level states $(a_{it}, \psi_{it}, \varphi_i)$ and product-specific factors.

When $\mathcal{B}_{it}(\hat{\theta}_{ipt}, \tau_{ipt}) < \mathcal{B}_{it}^d(\hat{\theta}_{ipt}, \tau_{ipt})$, the continuation value depends on whether the firm exits or continues to search. By assumption, both outcomes yield the same continuation value $\beta V^{ss}(\mathbf{0}; a_{it+1}, \psi_{it+1}, \varphi_i)$. Therefore, although the probability of exit depends on the evolution of match qualities in other product lines, this dependence does not affect the value of product line k , since exit and continuation are payoff-equivalent.

It follows that the value of product line k depends only on $(\hat{\theta}_{ipt}, \tau_{ipt}; a_{it}, \psi_{it}, \varphi_i)$ and not on Θ_{it} . This establishes independence across product lines. $\square$

Proof of Proposition 2, Learning Frictions and Output Wedges

Proof. Recall that the optimal output for product p by firm i with productivity φ_i , given beliefs $(\hat{\theta}_{ipt-1}, \tau_{ipt-1})$ in period t , is

$$y_{ipt}^* = \left(\frac{\eta - 1}{\eta} \right)^\eta P_{pt}^\eta C_{pt} \left[\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) \right]^\eta z(\varphi_i)^\eta w_t^{-\eta},$$

where $\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) = \exp \left\{ \frac{\hat{\theta}_{ipt-1}}{\eta} + \frac{1}{2} \left(\frac{\hat{\sigma}_{ipt-1}^2 + \sigma_\varepsilon^2}{\eta^2} \right) \right\}$, the posterior variance is given by $\hat{\sigma}_{ipt-1}^2 = \left(\frac{1}{\sigma_0^2} + \frac{\tau_{ipt-1}}{\sigma_\varepsilon^2} + \frac{\sum_{\tilde{t}} \psi_{i\tilde{t}}}{\sigma_\varepsilon^2} \right)^{-1}$, and the posterior mean can be written as

$$\hat{\theta}_{ipt-1} = \hat{\sigma}_{ipt-1}^2 \left[\frac{\theta_0}{\sigma_0^2} + \frac{\tau_{ipt-1} + \sum_{\tilde{t}} \psi_{i\tilde{t}} \log \theta_{ip}}{\sigma_\varepsilon^2} + \frac{1}{\sigma_\varepsilon^2} \left(\sum_{\tilde{t}} \varepsilon_{ip\tilde{t}} + \sum_{\tilde{t}} \psi_{i\tilde{t}} v_{ip\tilde{t}} \right) \right].$$

Taking expectations, we obtain the average value of $\left[\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) \right]^\eta$ as

$$\begin{aligned} \mathbb{E} \left[\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) \right]^\eta &= \exp \left[\hat{\sigma}_{ipt-1}^2 \left(\frac{\theta_0}{\sigma_0^2} + \frac{\tau_{ipt-1} + \sum_{\tilde{t}} \psi_{i\tilde{t}} \log \theta_{ip}}{\sigma_\varepsilon^2} \right) \right. \\ &\quad \left. + \frac{1}{2} (\hat{\sigma}_{ipt-1}^2)^2 \frac{\tau_{ipt-1} + \sum_{\tilde{t}} \psi_{i\tilde{t}}}{\sigma_\varepsilon^2} + \frac{1}{2} \frac{\hat{\sigma}_{ipt-1}^2 + \sigma_\varepsilon^2}{\eta} \right]. \end{aligned}$$

Let $A \equiv \frac{1}{\sigma_0^2}$ and $B \equiv \frac{\tau_{ipt-1} + \sum_{\tilde{t}} \psi_{i\tilde{t}}}{\sigma_\varepsilon^2}$. Then

$$\mathbb{E} \left[\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) \right]^\eta = \exp \left[\frac{A\theta_0 + B \log \theta_{ip}}{A+B} + \frac{B}{2(A+B)^2} + \frac{1}{2\eta} \left(\frac{1}{A+B} + \sigma_\varepsilon^2 \right) \right].$$

As $\psi_{it} \rightarrow \infty$ for all t , signals become perfectly informative and

$$\left[\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) \right]^\eta \rightarrow \theta_{ip},$$

even with a finite sample.

To characterize wedges, define the threshold

$$\theta_{ip}^* = \exp \left[\theta_0 + \frac{1}{2A} \left(\frac{B}{A+B} + \frac{1}{\eta} + \frac{A+B}{\eta} \sigma_\varepsilon^2 \right) \right].$$

Then, $\mathbb{E} \left[\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) \right]^\eta > \theta_{ip}$ if $\theta_{ip} < \theta_{ip}^*$, and $\mathbb{E} \left[\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) \right]^\eta < \theta_{ip}$ if $\theta_{ip} > \theta_{ip}^*$. Moreover, since $\frac{\partial \log \theta_{ip}^*}{\partial B} > 0$, firms with lower ψ_{it} have a lower threshold. Thus, low- ψ firms underestimate $\mathbb{E} \left[\mathcal{B}_{it-1}(\hat{\theta}_{ipt-1}, \tau_{ipt-1}) \right]^\eta$ for a wider range of high θ_{ip} .

Define the wedge

$$1 - \omega_{ipt} \equiv \frac{[\mathcal{B}_{it-1}]^\eta}{\theta_{ip}}.$$

Then output can be written as

$$y_{ipt}^* = (1 - \omega_{ipt}) \theta_{ip} z(\varphi_i)^\eta \left(\frac{\eta - 1}{\eta} \right)^\eta P_{pt}^\eta C_{pt} w_t^{-\eta}.$$

This implies that low- ψ firms underproduce high- θ_{ip} products and overproduce low- θ_{ip} products relative to the allocation implied by the true θ_{ip} . $\square$

Proof of Proposition 3, Biased Learning and Product Switching

Proof. The firm's problem can be characterized by three cases:

$$V_{i1}(\hat{\theta}_{ip1}, 1) = \begin{cases} -F^{ss} + \beta \left[x(\Xi_{i2}\mathcal{B}_{i1}(\hat{\theta}^0, 0)^\eta - \mathcal{F}^o) \right. \\ \quad \left. + (1-x)(\Xi_{i2}\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1)^\eta - \mathcal{F}^o) \right] & , \text{ if } \mathcal{B}_{i1}^{ss} \geq \mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1) > \mathcal{B}_{i1}^d \\ -F^{ss} + \beta x \left[\Xi_{i2}\mathcal{B}_{it}(\hat{\theta}^0, 0)^\eta - \mathcal{F}^o \right] & , \text{ if } \mathcal{B}_{i1}^d \geq \mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1) \\ \beta \left[\Xi_{i2}\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1)^\eta - \mathcal{F}^o \right] & , \text{ if } \mathcal{B}(\hat{\theta}_{ip1}, 1) > \mathcal{B}_{i1}^{ss}, \end{cases}$$

where $\Xi_{i2} \equiv \frac{(\eta-1)^{\eta-1}}{\eta^\eta} \left(\frac{z(\varphi_i)}{w_2} \right)^{\eta-1} P_{p2}^\eta C_{p2}$. The first two cases correspond to search, while the last corresponds to no search.

When $\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1) = \mathcal{B}_{i1}^{ss}$, the firm is indifferent between searching and not searching. Thus,

$$(\mathcal{B}_{i1}^{ss})^\eta = -\frac{\mathcal{F}^{ss}}{\beta x \Xi_{i2}} + \left(\mathcal{B}(\hat{\theta}^0, 0) \right)^\eta. \quad (\text{A.32})$$

Since Ξ_{i2} is increasing in productivity $z(\varphi_i)$, the search threshold $\mathcal{B}_{i1}^{ss}$ is increasing in $z(\varphi_i)$.

When $\mathcal{B}(\hat{\theta}_{ip1}, 1) = \mathcal{B}_{i1}^d$, the firm is indifferent between dropping and not retaining the product. Thus,

$$\mathcal{B}_{i1}^d = \left(\frac{\mathcal{F}^o}{\Xi_{i2}} \right)^{\frac{1}{\eta}}, \quad (\text{A.33})$$

which is a decreasing in productivity $z(\varphi_i)$.

Since

$$\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1) = \exp \left\{ \frac{\hat{\theta}_{ip1}}{\eta} + \frac{1}{2} \left(\frac{\hat{\sigma}_{ip1}^2 + \sigma_\varepsilon^2}{\eta^2} \right) \right\},$$

where

$$\hat{\theta}_{ip1} = \frac{\left(\frac{\theta_0}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} \log \theta_{ip1} + \frac{\psi_{i1} \log s_{ip1}}{\sigma_\varepsilon^2} \right)}{\left(\frac{1}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} + \frac{\psi_{i1}}{\sigma_\varepsilon^2} \right)}, \text{ and } \hat{\sigma}_{ip1}^2 = \frac{1}{\left(\frac{1}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} + \frac{\psi_{i1}}{\sigma_\varepsilon^2} \right)},$$

Substituting the expressions for $\hat{\theta}_{ip1}$ and $\hat{\sigma}_{ip1}^2$, we can rewrite

$$\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1) = \exp \left\{ \frac{1}{\eta} \left[\left(\frac{1}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} + \frac{\psi_{i1}}{\sigma_\varepsilon^2} \right)^{-1} \left(\frac{\theta_0}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} \log \theta_{ip1} + \frac{\psi_{i1} \log s_{ip1}}{\sigma_\varepsilon^2} + \frac{1}{2\eta} \right) + \frac{1}{2\eta} \sigma_\varepsilon^2 \right] \right\}.$$

Focusing on the terms involving ψ_{i1} , the derivative with respect to ψ_{i1} is proportional to

$$\left(\frac{1}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} \right) \log s_{ip1} - \left(\frac{\theta_0}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} \log \theta_{ip1} + \frac{1}{2\eta} \right),$$

where the proportionality factor is strictly positive. Since $\log \theta_{ipt} = \log \theta_{ip} + \varepsilon_{ipt}$, where $\varepsilon_{ipt} \sim N(0, \sigma_\varepsilon^2)$, and $\log s_{ipt} = \log \theta_{ip} + v_{ipt}$, where $v_{ipt} \sim N(0, \frac{\sigma_\varepsilon^2}{\psi_{it}})$, taking expectations with respect to the noise terms yields

$$\left(\frac{1}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} \right) \log \theta_{ip} - \left(\frac{\theta_0}{\sigma_0^2} + \frac{1}{\sigma_\varepsilon^2} \log \theta_{ip} + \frac{1}{2\eta} \right).$$

Therefore, for $\log \theta_{ip} > \theta_0 + \frac{\sigma_0^2}{2\eta}$, $\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1)$ is increasing in ψ_{i1} , whereas for $\log \theta_{ip} < \theta_0 + \frac{\sigma_0^2}{2\eta}$, it is decreasing in ψ_{i1} . This implies that, for a given realization of θ_{ip1} , tenure $\tau = 1$, and productivity, firms with lower ψ_{i1} are more likely to underestimate $\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1)$ when θ_{ip} is high. As a result, such firms are more likely to search for a new product ($\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1) < \mathcal{B}_{i1}^{ss}$) or even drop the product ($\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1) < \mathcal{B}_{i1}^d$), even though the true θ_{ip} is high enough to justify continued operation without search.

Conversely, when θ_{ip} is low, firms with lower ψ_{i1} are more likely to overestimate $\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1)$. In this case, they are more likely to continue operating the product, even though the true θ_{ip} would justify dropping it.

Furthermore, the likelihood that firms operate misaligned product lines for a given ψ_i gets higher if the firm has higher $z(\varphi_i)$. This is because, given all else equal, higher $z(\varphi_i)$ increases the search threshold $\mathcal{B}_{i1}^{ss}$ and decreases the drop threshold $\mathcal{B}_{i1}^d$ decreases in $z(\varphi_i)$, which widens the range of posterior mean $\mathcal{B}_{i1}(\hat{\theta}_{ip1}, 1)$, over which firms both continue operating and search. This expanded region induces more firms—even those with relatively precise beliefs (high ψ_i)—to switch away from high- θ_{ip} products or retain

low- θ_{ip} products. As a result, misallocation driven by learning frictions becomes more pronounced among high-productivity firms. $\square$

B Empiric

B.1 Product Dropping

Table A1: Probability of Product Drop (PV)

VARIABLES	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$
$\log(PV_{pit})$	-0.055*** (0.002)	-0.054*** (0.002)	-0.053*** (0.002)	-0.053*** (0.002)	-0.052*** (0.002)
$\log(\text{Emp})$			-0.020*** (0.002)	-0.009*** (0.002)	-0.012*** (0.002)
$\log(\text{Total Assets})$			0.003*** (0.001)	0.019*** (0.004)	0.019*** (0.004)
# establishments			-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
# products			0.007*** (0.000)	0.007*** (0.000)	0.006*** (0.000)
$\log(\text{CA})$			-0.002*** (0.001)	-0.002*** (0.001)	-0.002*** (0.001)
$\log(\text{CS})$			-0.004*** (0.001)	-0.004*** (0.001)	-0.003*** (0.001)
TFP			-0.004* (0.002)	-0.006** (0.002)	-0.007*** (0.003)
$\log(\text{Prod. Emp})$				-0.009*** (0.001)	-0.010*** (0.001)
$\log(\text{Equipment})$				-0.028*** (0.004)	-0.029*** (0.004)
$\log(\text{Structure})$				0.011*** (0.003)	0.011*** (0.003)
$\log(\text{Material})$				-0.004*** (0.001)	-0.004*** (0.001)
$\log(\text{Energy})$				0.004*** (0.001)	0.003*** (0.001)
Tenure		0.004*** (0.000)			0.004*** (0.001)
Tenure squared		-0.000*** (0.000)			-0.000*** (0.000)
Constant	0.943*** (0.010)	0.942*** (0.010)	0.965*** (0.015)	0.978*** (0.016)	1.004*** (0.017)

Notes: Estimates for product-year, firm fixed effects, and the constant are suppressed. Observations are unweighted. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A2: Probability of Product Drop (Match Quality)

VARIABLES	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$
Quality _{pi}	-1.212*** (0.029)	-1.028*** (0.022)	-1.189*** (0.029)	-1.187*** (0.029)	-1.074*** (0.021)
log(Emp)			-0.035*** (0.002)	-0.015*** (0.002)	-0.016*** (0.002)
log(Total Assets)			0.002* (0.001)	0.023*** (0.004)	0.021*** (0.004)
# establishments			-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
# products			0.005*** (0.000)	0.005*** (0.000)	0.005*** (0.000)
log(CA)			0 (0.001)	0 (0.001)	0.002*** (0.001)
log(CS)			-0.005*** (0.001)	-0.005*** (0.001)	-0.002*** (0.001)
TFP			-0.011*** (0.002)	-0.020*** (0.002)	-0.013*** (0.002)
log(Prod. Emp)				-0.009*** (0.001)	-0.010*** (0.001)
log(Equipment)				-0.025*** (0.004)	-0.025*** (0.004)
log(Structure)				0.005** (0.003)	0.006** (0.003)
log(Material)				-0.013*** (0.001)	-0.008*** (0.001)
log(Energy)				0.001 (0.001)	0.002* (0.001)
log(PV _{pit})		-0.021*** (0.001)			-0.017*** (0.001)
Tenure		0.041*** (0.001)			0.042*** (0.001)
Tenure squared		-0.001*** (0.000)			-0.001*** (0.000)
Constant	1.053*** (0.015)	0.963*** (0.014)	1.164*** (0.021)	1.221*** (0.022)	1.111*** (0.022)

Notes: Estimates for product-year, firm fixed effects, and the constant are suppressed. Observations are unweighted. * p<0.1, ** p<0.05, *** p<0.01.

Table A3: Probability of Product Drop (Young Firm)

VARIABLES	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$
Young Firm _{it}	-0.084*** (0.004)	-0.073*** (0.003)	-0.092*** (0.004)	-0.093*** (0.004)	-0.075*** (0.003)
log(Emp)			-0.050*** (0.002)	-0.025*** (0.002)	-0.015*** (0.002)
log(Total Assets)			-0.008*** (0.001)	0.015*** (0.005)	0.019*** (0.004)
# establishments			-0.004*** (0.000)	-0.004*** (0.000)	-0.002*** (0.000)
# products			0.011*** (0.001)	0.011*** (0.001)	0.006*** (0.000)
log(CA)			-0.003*** (0.001)	-0.002*** (0.001)	-0.001** (0.001)
log(CS)			-0.002** (0.001)	-0.002** (0.001)	-0.003*** (0.001)
TFP			-0.014*** (0.002)	-0.027*** (0.003)	-0.008*** (0.003)
log(Prod. Emp)				-0.008*** (0.001)	-0.010*** (0.001)
log(Equipment)				-0.032*** (0.004)	-0.029*** (0.004)
log(Structure)				0.011*** (0.003)	0.011*** (0.003)
log(Material)				-0.019*** (0.001)	-0.005*** (0.001)
log(Energy)				-0.002* (0.001)	0.003*** (0.001)
log(PV _{pit})		-0.054*** (0.002)			-0.052*** (0.002)
Tenure		0.002*** (0.000)			0.003*** (0.000)
Tenure squared		-0.000*** (0.000)			-0.000*** (0.000)
Constant	0.591*** (0.000)	0.951*** (0.011)	0.828*** (0.015)	0.918*** (0.016)	1.038*** (0.018)

Notes: Estimates for product-year, firm fixed effects, and the constant are suppressed. Observations are unweighted. * p<0.1, ** p<0.05, *** p<0.01.

Table A4: Probability of Product Drop ($PV \times \text{Young}$)

VARIABLES	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$
$\log(PV_{pit})$	-0.056*** (0.002)	-0.055*** (0.002)	-0.054*** (0.002)	-0.054*** (0.002)	-0.052*** (0.002)
Young Firm _{it}	-0.163*** (0.008)	-0.142*** (0.007)	-0.159*** (0.008)	-0.160*** (0.008)	-0.141*** (0.007)
$\log(PV_{pit}) \times \text{Young}_{it}$	0.012*** (0.001)	0.011*** (0.001)	0.011*** (0.001)	0.011*** (0.001)	0.010*** (0.001)
$\log(\text{Emp})$			-0.025*** (0.002)	-0.013*** (0.002)	-0.015*** (0.002)
$\log(\text{Total Assets})$			0.001 (0.001)	0.019*** (0.004)	0.019*** (0.004)
# establishments			-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
# products			0.007*** (0.000)	0.007*** (0.000)	0.006*** (0.000)
$\log(\text{CA})$			-0.002*** (0.001)	-0.001* (0.001)	-0.001* (0.001)
$\log(\text{CS})$			-0.003*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)
TFP			-0.005** (0.002)	-0.008*** (0.003)	-0.009*** (0.003)
$\log(\text{Prod. Emp})$				-0.009*** (0.001)	-0.010*** (0.001)
$\log(\text{Equipment})$				-0.029*** (0.004)	-0.029*** (0.004)
$\log(\text{Structure})$				0.011*** (0.003)	0.011*** (0.003)
$\log(\text{Material})$				-0.004*** (0.001)	-0.005*** (0.001)
$\log(\text{Energy})$				0.004*** (0.001)	0.003*** (0.001)
Tenure		0.003*** (0.000)			0.003*** (0.000)
Tenure squared		-0.000*** (0.000)			-0.000*** (0.000)
Constant	0.960*** (0.011)	0.957*** (0.011)	1.012*** (0.016)	1.027*** (0.017)	1.044*** (0.018)

Notes: Estimates for product-year, firm fixed effects, and the constant are suppressed. Observations are unweighted. * p<0.1, ** p<0.05, *** p<0.01.

Table A5: Probability of Product Drop (Quality $\times$ Young)

VARIABLES	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$	$Drop_{t:t+5}$
Quality _{<i>pi</i>}	-1.221*** (0.030)	-1.032*** (0.022)	-1.196*** (0.030)	-1.194*** (0.030)	-1.081*** (0.022)
Young Firm _{<i>it</i>}	-0.411*** (0.022)	-0.371*** (0.023)	-0.402*** (0.022)	-0.403*** (0.022)	-0.368*** (0.022)
Quality _{<i>pi</i>} $\times$ Young _{<i>it</i>}	0.395*** (0.044)	0.514*** (0.043)	0.367*** (0.043)	0.368*** (0.043)	0.497*** (0.043)
log(Emp)			-0.040*** (0.002)	-0.020*** (0.002)	-0.018*** (0.002)
log(Total Assets)			0 (0.001)	0.022*** (0.004)	0.020*** (0.004)
# establishments			-0.002*** (0.000)	-0.002*** (0.000)	-0.001*** (0.000)
# products			0.005*** (0.000)	0.005*** (0.000)	0.005*** (0.000)
log(CA)			0 (0.001)	0.001* (0.001)	0.002*** (0.001)
log(CS)			-0.004*** (0.001)	-0.004*** (0.001)	-0.002*** (0.001)
TFP			-0.012*** (0.002)	-0.021*** (0.002)	-0.014*** (0.002)
log(Prod. Emp)				-0.009*** (0.001)	-0.010*** (0.001)
log(Equipment)				-0.026*** (0.004)	-0.026*** (0.004)
log(Structure)				0.005* (0.003)	0.006** (0.003)
log(Material)				-0.014*** (0.001)	-0.008*** (0.001)
log(Energy)				0 (0.001)	0.002* (0.001)
log(PV_{pit})		-0.021*** (0.001)			-0.017*** (0.001)
Tenure		0.039*** (0.001)			0.039*** (0.001)
Tenure squared		-0.001*** (0.000)			-0.001*** (0.000)
Constant	1.071*** (0.015)	0.980*** (0.014)	1.222*** (0.021)	1.280*** (0.023)	1.146*** (0.022)

Notes: Estimates for product-year, firm fixed effects, and the constant are suppressed. Observations are unweighted. * p<0.1, ** p<0.05, *** p<0.01.

B.2 Firm-Level Evidence

Table A6: Product Add and Drop

	$Add_{t:t+5}$	$Add_{t:t+5}$	$Add_{t:t+5}$
$Drop_{t:t+5}$	0.096*** (0.012)		0.070*** (0.012)
Young Firm _{it}		0.148*** (0.005)	0 (0.005)
$Drop_{t:t+5} \times \text{Young}_{it}$			0.295*** (0.008)
log(Emp)	0.022*** (0.002)	0.029*** (0.002)	0.030*** (0.002)
log(Total Assets)	-0.007* (0.004)	-0.005 (0.004)	-0.006* (0.004)
# establishments	0.015*** (0.004)	0.012*** (0.003)	0.014*** (0.004)
# products	-0.034*** (0.003)	-0.027*** (0.002)	-0.034*** (0.003)
log(CA)	-0.001 (0.001)	-0.003** (0.001)	-0.003** (0.001)
log(CS)	-0.005*** (0.001)	-0.007*** (0.001)	-0.007*** (0.001)
TFP	0.007*** (0.003)	0.008*** (0.003)	0.010*** (0.003)
log(Prod. Emp)	-0.006*** (0.001)	-0.008*** (0.001)	-0.005*** (0.001)
log(Equipment)	0.022*** (0.003)	0.021*** (0.003)	0.023*** (0.003)
log(Structure)	-0.010*** (0.003)	-0.008*** (0.003)	-0.008*** (0.003)
log(Material)	0.010*** (0.001)	0.010*** (0.001)	0.011*** (0.001)
log(Energy)	0.006*** (0.001)	0.006*** (0.001)	0.006*** (0.001)
Constant	0.172*** (0.018)	0.174*** (0.018)	0.116*** (0.018)
Observations	682,000	682,000	682,000
Fixed effects	i, jt	i, jt	i, jt

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Observations are unweighted. * p<0.1, ** p<0.05, *** p<0.01.

Table A7: Product Add and Drop (Baseline)

	$Add_{t:t+5}$	$Add_{t:t+5}$	$Add_{t:t+5}$
$Drop_{t:t+5}$	0.066*** (0.014)	0.095*** (0.013)	0.096*** (0.012)
log(Emp)		0.027*** (0.002)	0.022*** (0.002)
log(Total Assets)		0.008*** (0.001)	-0.007* (0.004)
# establishments		0.015*** (0.004)	0.015*** (0.004)
# products		-0.034*** (0.003)	-0.034*** (0.003)
log(CA)		-0.001 (0.001)	-0.001 (0.001)
log(CS)		-0.005*** (0.001)	-0.005*** (0.001)
TFP		-0.002 (0.002)	0.007*** (0.003)
log(Prod. Emp)			-0.006*** (0.001)
log(Equipment)			0.022*** (0.003)
log(Structure)			-0.010*** (0.003)
log(Material)			0.010*** (0.001)
log(Energy)			0.006*** (0.001)
Constant	0.325*** (0.008)	0.221*** (0.015)	0.172*** (0.018)

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Observations are unweighted. * p<0.1, ** p<0.05, *** p<0.01.

Table A8: Product Add and Drop (Young Firm)

	$Add_{t:t+5}$	$Add_{t:t+5}$	$Add_{t:t+5}$
Young Firm _{it}	0.140*** (0.005)	0.148*** (0.005)	0.148*** (0.005)
log(Emp)		0.032*** (0.003)	0.029*** (0.002)
log(Total Assets)		0.010*** (0.001)	-0.005 (0.004)
# establishments		0.012*** (0.004)	0.012*** (0.003)
# products		-0.027*** (0.002)	-0.027*** (0.002)
log(CA)		-0.003*** (0.001)	-0.003** (0.001)
log(CS)		-0.006*** (0.001)	-0.007*** (0.001)
TFP		-0.001 (0.003)	0.008*** (0.003)
log(Prod. Emp)			-0.008*** (0.001)
log(Equipment)			0.021*** (0.003)
log(Structure)			-0.008*** (0.003)
log(Material)			0.010*** (0.001)
log(Energy)			0.006*** (0.001)
Constant	0.350*** (0.002)	0.220*** (0.015)	0.174*** (0.018)

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Observations are unweighted. * p<0.1, ** p<0.05, *** p<0.01.

Table A9: Product Add and Drop (Drop $\times$ Young)

	$Add_{t:t+5}$	$Add_{t:t+5}$	$Add_{t:t+5}$
$Drop_{t:t+5}$	0.039*** (0.013)	0.069*** (0.012)	0.070*** (0.012)
Young Firm $_{it}$	-0.015*** (0.005)	-0.001 (0.005)	0 (0.005)
$Drop_{t:t+5} \times Young_{it}$	0.303*** (0.008)	0.295*** (0.008)	0.295*** (0.008)
log(Emp)		0.037*** (0.002)	0.030*** (0.002)
log(Total Assets)		0.011*** (0.001)	-0.006* (0.004)
# establishments		0.014*** (0.004)	0.014*** (0.004)
# products		-0.034*** (0.003)	-0.034*** (0.003)
log(CA)		-0.003** (0.001)	-0.003** (0.001)
log(CS)		-0.006*** (0.001)	-0.007*** (0.001)
TFP		0 (0.002)	0.010*** (0.003)
log(Prod. Emp)			-0.005*** (0.001)
log(Equipment)			0.023*** (0.003)
log(Structure)			-0.008*** (0.003)
log(Material)			0.011*** (0.001)
log(Energy)			0.006*** (0.001)
Constant	0.325*** (0.008)	0.168*** (0.016)	0.116*** (0.018)

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Observations are unweighted. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A10: Number of Products Added and Drop

	$nAdd_{t:t+5}$	$nAdd_{t:t+5}$	$nAdd_{t:t+5}$
$Drop_{t:t+5}$	0.130*** (0.026)		0.086*** (0.025)
Young Firm _{it}		0.206*** (0.014)	-0.044*** (0.012)
$Drop_{t:t+5} \times \text{Young}_{it}$			0.490*** (0.024)
log(Emp)	0.071*** (0.009)	0.081*** (0.009)	0.082*** (0.009)
log(Total Assets)	-0.006 (0.008)	-0.004 (0.007)	-0.006 (0.007)
# establishments	0.077*** (0.026)	0.074*** (0.025)	0.077*** (0.026)
# products	-0.150*** (0.022)	-0.140*** (0.021)	-0.149*** (0.022)
log(CA)	-0.039*** (0.005)	-0.041*** (0.005)	-0.041*** (0.005)
log(CS)	-0.070*** (0.007)	-0.071*** (0.007)	-0.071*** (0.007)
TFP	0.014* (0.008)	0.015* (0.008)	0.017** (0.008)
log(Prod. Emp)	-0.013*** (0.003)	-0.015*** (0.004)	-0.011*** (0.003)
log(Equipment)	0.028*** (0.007)	0.027*** (0.007)	0.029*** (0.007)
log(Structure)	-0.01 (0.007)	-0.008 (0.007)	-0.008 (0.007)
log(Material)	0.017*** (0.003)	0.016*** (0.003)	0.018*** (0.003)
log(Energy)	0.016*** (0.004)	0.016*** (0.004)	0.016*** (0.004)
Constant	0.393*** (0.042)	0.394*** (0.047)	0.324*** (0.044)
Observations	682,000	682,000	682,000
Fixed effects	i, jt	i, jt	i, jt
Controls	Full	Full	Full

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Observations are unweighted. * p<0.1, ** p<0.05, *** p<0.01.

Table A11: Relative Match Quality of Added Products

	$Add_{t:t+5}^{high}$	$Add_{t:t+5}^{high}$	$Add_{t:t+5}^{high}$
$Drop_{t:t+5}$	0.028** (0.011)		0.049*** (0.010)
Young Firm _{it}		-0.112*** (0.017)	0.044* (0.024)
$Drop_{t:t+5} \times \text{Young}_{it}$			-0.228*** (0.024)
log(Emp)	-0.011 (0.009)	-0.018** (0.009)	-0.019** (0.009)
log(Total Assets)	0.039*** (0.012)	0.037*** (0.012)	0.037*** (0.012)
# establishments	-0.007*** (0.003)	-0.007*** (0.003)	-0.007*** (0.003)
# products	0.019*** (0.003)	0.020*** (0.003)	0.019*** (0.003)
log(CA)	0.001 (0.003)	0.002 (0.003)	0.001 (0.003)
log(CS)	-0.005 (0.003)	-0.003 (0.003)	-0.003 (0.003)
TFP	-0.031*** (0.009)	-0.034*** (0.009)	-0.034*** (0.009)
log(Prod. Emp)	0.022*** (0.003)	0.021*** (0.003)	0.021*** (0.003)
log(Equipment)	-0.025*** (0.009)	-0.025*** (0.009)	-0.025*** (0.009)
log(Structure)	-0.016* (0.009)	-0.015* (0.009)	-0.015* (0.009)
log(Material)	-0.015*** (0.005)	-0.016*** (0.004)	-0.017*** (0.004)
log(Energy)	-0.014*** (0.005)	-0.013*** (0.005)	-0.014*** (0.005)
Constant	0.430*** (0.050)	0.516*** (0.049)	0.487*** (0.048)
Observations	66,000	66,000	66,000
Fixed effects	i, jt	i, jt	i, jt

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Robust standard errors clustered at the firm's modal five-digit SIC product level are reported in parentheses. Observations are unweighted, and observation counts are rounded due to the Census Bureau disclosure avoidance procedures. * p<0.1, ** p<0.05, *** p<0.01.

Table A12: Relative Match Quality of Added Products (Drop)

	$Add_{t:t+5}^{high}$	$Add_{t:t+5}^{high}$	$Add_{t:t+5}^{high}$
$Drop_{t:t+5}$	0.052*** (0.011)	0.026** (0.011)	0.028** (0.011)
log(Emp)		-0.008 (0.008)	-0.011 (0.009)
log(Total Assets)		-0.006 (0.005)	0.039*** (0.012)
# establishments		-0.007*** (0.003)	-0.007*** (0.003)
# products		0.020*** (0.003)	0.019*** (0.003)
log(CA)		0.002 (0.003)	0.001 (0.003)
log(CS)		-0.005 (0.003)	-0.005 (0.003)
TFP		-0.018** (0.008)	-0.031*** (0.009)
log(Prod. Emp)			0.022*** (0.003)
log(Equipment)			-0.025*** (0.009)
log(Structure)			-0.016* (0.009)
log(Material)			-0.015*** (0.005)
log(Energy)			-0.014*** (0.005)
Constant	0.301*** (0.007)	0.382*** (0.041)	0.430*** (0.050)

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Robust standard errors clustered at the firm's modal five-digit SIC product level are reported in parentheses. Observations are unweighted, and observation counts are rounded due to the Census Bureau disclosure avoidance procedures. * p<0.1, ** p<0.05, *** p<0.01.

Table A13: Relative Match Quality of Added Products (Young Firm)

	$Add_{t:t+5}^{high}$	$Add_{t:t+5}^{high}$	$Add_{t:t+5}^{high}$
Young Firm _{it}	-0.115*** (0.016)	-0.114*** (0.017)	-0.112*** (0.017)
log(Emp)		-0.018** (0.008)	-0.018** (0.009)
log(Total Assets)		-0.007 (0.005)	0.037*** (0.012)
# establishments		-0.007*** (0.003)	-0.007*** (0.003)
# products		0.020*** (0.003)	0.020*** (0.003)
log(CA)		0.003 (0.003)	0.002 (0.003)
log(CS)		-0.004 (0.003)	-0.003 (0.003)
TFP		-0.020** (0.008)	-0.034*** (0.009)
log(Prod. Emp)			0.021*** (0.003)
log(Equipment)			-0.025*** (0.009)
log(Structure)			-0.015* (0.009)
log(Material)			-0.016*** (0.004)
log(Energy)			-0.013*** (0.005)
Constant	0.347*** (0.003)	0.463*** (0.041)	0.516*** (0.049)

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Robust standard errors clustered at the firm's modal five-digit SIC product level are reported in parentheses. Observations are unweighted, and observation counts are rounded due to the Census Bureau disclosure avoidance procedures. * p<0.1, ** p<0.05, *** p<0.01.

Table A14: Relative Match Quality of Added Products (Drop $\times$ Young)

	$Add_{t:t+5}^{high}$	$Add_{t:t+5}^{high}$	$Add_{t:t+5}^{high}$
$Drop_{t:t+5}$	0.073*** (0.010)	0.047*** (0.010)	0.049*** (0.010)
Young Firm $_{it}$	0.043* (0.023)	0.042* (0.024)	0.044* (0.024)
$Drop_{t:t+5} \times Young_{it}$	-0.231*** (0.025)	-0.227*** (0.024)	-0.228*** (0.024)
log(Emp)		-0.019** (0.008)	-0.019** (0.009)
log(Total Assets)		-0.007 (0.005)	0.037*** (0.012)
# establishments		-0.007*** (0.003)	-0.007*** (0.003)
# products		0.019*** (0.003)	0.019*** (0.003)
log(CA)		0.003 (0.003)	0.001 (0.003)
log(CS)		-0.004 (0.003)	-0.003 (0.003)
TFP		-0.020** (0.008)	-0.034*** (0.009)
log(Prod. Emp)			0.021*** (0.003)
log(Equipment)			-0.025*** (0.009)
log(Structure)			-0.015* (0.009)
log(Material)			-0.017*** (0.004)
log(Energy)			-0.014*** (0.005)
Constant	0.295*** (0.007)	0.433*** (0.041)	0.487*** (0.048)

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Robust standard errors clustered at the firm's modal five-digit SIC product level are reported in parentheses. Observations are unweighted, and observation counts are rounded due to the Census Bureau disclosure avoidance procedures. * p<0.1, ** p<0.05, *** p<0.01.

Table A15: Material-Based Closeness of Added Products

	$Mat_{t:t+5}^{add}$	$Mat_{t:t+5}^{add}$	$Mat_{t:t+5}^{add}$
$Drop_{t:t+5}$	0.098*** (0.004)		0.102*** (0.004)
Young Firm _{it}		-0.003 (0.004)	0.022*** (0.005)
$Drop_{t:t+5} \times \text{Young}_{it}$			-0.037*** (0.006)
log(Emp)	-0.002 (0.003)	-0.003 (0.003)	-0.003 (0.003)
log(Total Assets)	-0.001 (0.004)	-0.001 (0.004)	-0.001 (0.004)
# establishments	0.006*** (0.002)	0.005*** (0.002)	0.006*** (0.002)
# products	-0.028*** (0.003)	-0.024*** (0.002)	-0.028*** (0.003)
log(CA)	-0.003*** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)
log(CS)	-0.007*** (0.001)	-0.007*** (0.001)	-0.007*** (0.001)
TFP	-0.008*** (0.002)	-0.007*** (0.002)	-0.008*** (0.002)
# materials	0.004*** (0.000)	0.005*** (0.000)	0.004*** (0.000)
log(Prod. Emp)	0 (0.001)	-0.001 (0.001)	0 (0.001)
log(Equipment)	-0.004* (0.003)	-0.005* (0.003)	-0.005* (0.003)
log(Structure)	0.004 (0.002)	0.004* (0.002)	0.003 (0.002)
log(Material)	-0.005*** (0.001)	-0.005*** (0.001)	-0.005*** (0.001)
log(Energy)	-0.007*** (0.001)	-0.007*** (0.001)	-0.007*** (0.001)
Constant	0.458*** (0.013)	0.521*** (0.013)	0.457*** (0.013)
Observations	198,000	198,000	198,000
Fixed effects	i, jt	i, jt	i, jt

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Robust standard errors clustered at the firm's modal five-digit SIC product level are reported in parentheses. Observations are unweighted, and observation counts are rounded due to the Census Bureau disclosure avoidance procedures. * p<0.1, ** p<0.05, *** p<0.01.

Table A16: Material-Based Closeness of Added Products (Drop)

	$Mat_{t:t+5}^{add}$	$Mat_{t:t+5}^{add}$	$Mat_{t:t+5}^{add}$
$Drop_{t:t+5}$	0.093*** (0.007)	0.098*** (0.004)	0.098*** (0.004)
log(Emp)		-0.010*** (0.002)	-0.002 (0.003)
log(Total Assets)		-0.004*** (0.001)	-0.001 (0.004)
# establishments		0.006*** (0.002)	0.006*** (0.002)
# products		-0.028*** (0.003)	-0.028*** (0.003)
log(CA)		-0.003*** (0.001)	-0.003*** (0.001)
log(CS)		-0.007*** (0.001)	-0.007*** (0.001)
TFP		-0.003 (0.002)	-0.008*** (0.002)
# materials		0.004*** (0.000)	0.004*** (0.000)
log(Prod. Emp)			0 (0.001)
log(Equipment)			-0.004* (0.003)
log(Structure)			0.004 (0.002)
log(Material)			-0.005*** (0.001)
log(Energy)			-0.007*** (0.001)
Constant	0.454*** (0.006)	0.427*** (0.010)	0.458*** (0.013)
Observations	198,000	198,000	198,000
Fixed effects	i, jt	i, jt	i, jt

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Robust standard errors clustered at the firm's modal five-digit SIC product level are reported in parentheses. Observations are unweighted, and observation counts are rounded due to the Census Bureau disclosure avoidance procedures. * p<0.1, ** p<0.05, *** p<0.01.

Table A17: Material-Based Closeness of Added Products (Young Firm)

	$Mat_{t:t+5}^{add}$	$Mat_{t:t+5}^{add}$	$Mat_{t:t+5}^{add}$
Young Firm _{it}	-0.005 (0.005)	-0.003 (0.004)	-0.003 (0.004)
log(Emp)		-0.012*** (0.002)	-0.003 (0.003)
log(Total Assets)		-0.004*** (0.001)	-0.001 (0.004)
# establishments		0.005*** (0.002)	0.005*** (0.002)
# products		-0.024*** (0.002)	-0.024*** (0.002)
log(CA)		-0.003*** (0.001)	-0.003*** (0.001)
log(CS)		-0.007*** (0.001)	-0.007*** (0.001)
TFP		-0.003 (0.002)	-0.007*** (0.002)
# materials		0.005*** (0.000)	0.005*** (0.000)
log(Prod. Emp)			-0.001 (0.001)
log(Equipment)			-0.005* (0.003)
log(Structure)			0.004* (0.002)
log(Material)			-0.005*** (0.001)
log(Energy)			-0.007*** (0.001)
Constant	0.520*** (0.003)	0.489*** (0.011)	0.521*** (0.013)
Observations	198,000	198,000	198,000
Fixed effects	<i>i, jt</i>	<i>i, jt</i>	<i>i, jt</i>

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Robust standard errors clustered at the firm's modal five-digit SIC product level are reported in parentheses. Observations are unweighted, and observation counts are rounded due to the Census Bureau disclosure avoidance procedures. * p<0.1, ** p<0.05, *** p<0.01.

Table A18: Material-Based Closeness of Added Products (Drop $\times$ Young)

	$Mat_{t:t+5}^{add}$	$Mat_{t:t+5}^{add}$	$Mat_{t:t+5}^{add}$
$Drop_{t:t+5}$	0.097*** (0.007)	0.101*** (0.004)	0.102*** (0.004)
Young Firm _{it}	0.028*** (0.006)	0.023*** (0.005)	0.022*** (0.005)
$Drop_{t:t+5} \times Young_{it}$	-0.046*** (0.006)	-0.037*** (0.006)	-0.037*** (0.006)
log(Emp)		-0.010*** (0.002)	-0.003 (0.003)
log(Total Assets)		-0.004*** (0.001)	-0.001 (0.004)
# establishments		0.006*** (0.002)	0.006*** (0.002)
# products		-0.028*** (0.003)	-0.028*** (0.003)
log(CA)		-0.003*** (0.001)	-0.003*** (0.001)
log(CS)		-0.007*** (0.001)	-0.007*** (0.001)
TFP		-0.003 (0.002)	-0.008*** (0.002)
# materials		0.004*** (0.000)	0.004*** (0.000)
log(Prod. Emp)			0 (0.001)
log(Equipment)			-0.005* (0.003)
log(Structure)			0.003 (0.002)
log(Material)			-0.005*** (0.001)
log(Energy)			-0.007*** (0.001)
Constant	0.451*** (0.006)	0.426*** (0.010)	0.457*** (0.013)
Observations	198,000	198,000	198,000
Fixed effects	i, jt	i, jt	i, jt

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Robust standard errors clustered at the firm's modal five-digit SIC product level are reported in parentheses. Observations are unweighted, and observation counts are rounded due to the Census Bureau disclosure avoidance procedures. * p<0.1, ** p<0.05, *** p<0.01.

Table A19: Industry Closeness of Added Products

	$Ind4_{t:t+5}^{add}$	$Ind4_{t:t+5}^{add}$	$Ind4_{t:t+5}^{add}$
$Drop_{t:t+5}$	0.079*** (0.006)		0.081*** (0.006)
Young Firm _{it}		0.018*** (0.006)	0.032*** (0.008)
$Drop_{t:t+5} \times \text{Young}_{it}$			-0.019** (0.009)
log(Emp)	0.010*** (0.004)	0.012*** (0.004)	0.012*** (0.004)
log(Total Assets)	-0.004 (0.005)	-0.004 (0.005)	-0.003 (0.005)
# establishments	0.003** (0.001)	0.001 (0.001)	0.003** (0.001)
# products	-0.007*** (0.002)	-0.003* (0.002)	-0.007*** (0.002)
log(CA)	0.002* (0.001)	0.002 (0.001)	0.002 (0.001)
log(CS)	0.003** (0.001)	0.003** (0.001)	0.003** (0.001)
TFP	-0.005 (0.003)	-0.004 (0.003)	-0.004 (0.003)
log(Prod. Emp)	-0.009*** (0.002)	-0.011*** (0.002)	-0.009*** (0.002)
log(Equipment)	0.001 (0.004)	0.001 (0.004)	0.001 (0.004)
log(Structure)	0.004 (0.003)	0.004 (0.003)	0.004 (0.003)
log(Material)	0.001 (0.002)	0.002 (0.002)	0.002 (0.002)
log(Energy)	0.003 (0.002)	0.003* (0.002)	0.003 (0.002)
Constant	0.213*** (0.016)	0.253*** (0.017)	0.200*** (0.017)
Observations	198,000	198,000	198,000
Fixed effects	i, jt	i, jt	i, jt

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Robust standard errors clustered at the firm's modal five-digit SIC product level are reported in parentheses. Observations are unweighted, and observation counts are rounded due to the Census Bureau disclosure avoidance procedures. * p<0.1, ** p<0.05, *** p<0.01.

Table A20: Product Add/Drop and Firm Performance

	ΔV_{add}	ΔEmp	ΔTFP	ΔLP
$Add_{t:t+5}$	0.830*** (0.019)	0.802*** (0.019)	0.735*** (0.019)	0.762*** (0.019)
$Drop_{t:t+5}$	-0.754*** (0.019)	-0.731*** (0.018)	-0.686*** (0.017)	-0.713*** (0.018)
log(Emp)	-0.187*** (0.009)	-0.604*** (0.008)	0.099*** (0.007)	0.200*** (0.007)
log(Total Assets)	-0.071*** (0.006)	-0.021*** (0.006)	-0.026** (0.012)	-0.018 (0.011)
# establishments	-0.027*** (0.008)	-0.024*** (0.007)	-0.026*** (0.008)	-0.028*** (0.008)
# products	0.077*** (0.006)	0.079*** (0.005)	0.070*** (0.005)	0.072*** (0.006)
log(CA)	-0.003* (0.002)	0.006*** (0.002)	0.002 (0.002)	0.003 (0.002)
log(CS)	0.014*** (0.002)	0.014*** (0.002)	0.011*** (0.002)	0.012*** (0.002)
TFP	-0.561*** (0.012)	0.130*** (0.006)	-0.369*** (0.022)	-0.088*** (0.026)
log(Prod. Emp)	0.021*** (0.003)	0.028*** (0.003)	0.019*** (0.003)	0.001 (0.003)
log(Equipment)	0.030*** (0.005)	0.075*** (0.005)	0.059*** (0.007)	0.029*** (0.006)
log(Structure)	-0.046*** (0.005)	-0.028*** (0.005)	-0.028*** (0.008)	-0.035*** (0.007)
log(Material)	-0.049*** (0.006)	0.068*** (0.003)	0.023*** (0.006)	-0.042*** (0.008)
log(Energy)	-0.030*** (0.004)	0.042*** (0.002)	0.010*** (0.003)	0.035*** (0.004)
Constant	2.252*** (0.052)	0.424*** (0.047)	-0.415*** (0.081)	-0.633*** (0.100)
Observations	682,000	682,000	648,000	648,000
Fixed effects	i, jt	i, jt	i, jt	i, jt
Controls	Full	Full	Full	Full

Notes: Estimates for industry-year, firm fixed effects, and the constant are suppressed. Robust standard errors clustered at the firm's modal five-digit SIC product level are reported in parentheses. Observations are unweighted, and observation counts are rounded due to the Census Bureau disclosure avoidance procedures. * p<0.1, ** p<0.05, *** p<0.01.

C Alternative Match Quality Estimation

One alternation of the quality estimation is to use product-year and firm-year fixed effects as follows, instead of controlling for the time-varying firm and industry characteristics:

$$y_{ipjt} = \theta_{ip} + X_{ipjt}\gamma + \delta_{pt} + \delta_{it} + \varepsilon_{ipjt}, \quad (\text{C.34})$$

where δ_{pt} is a product-year fixed effect, and δ_{it} is a firm-year fixed effect.

δ_{pt} controls for product specific characteristics as well as product-year specific shocks. Again, this is because each product is subject to different production technology or demand structure. Therefore, it is possible that there are several other effects affecting the value of shipment, attributed to product-year specific technology or demand shocks.

In a similar fashion, firms or entrepreneurs have different base sets of available resources or ability to manufacture products (i.e. a financing constraint, customer capital, brand values, and marketing resources, etc.). The value of shipment for each product can also be influenced by these firm-level properties. Furthermore, any types of industry-specific characteristics that can give a substantial impact on the total value of shipment are absorbed by the firm-year fixed effect. Therefore, equation (C.34) could be one alternative of the baseline quality estimation in (4.22).

However, estimation based on (C.34) faces several limitations due to the inclusion of multiple fixed effects. In particular, estimating firm-product and firm-year fixed effects simultaneously leads to the exclusion of a non-negligible portion of the sample, as some products appear in only one firm or some firms produce only a single product in a given year. Therefore, an alternative approach is to replace the firm-year fixed effect δ_{it} with an extensive set of firm-level controls X_{ijt} , which is our baseline specification (4.22).

Alternatively, we could implement an AKM-style specification (Abowd et al., 1999) and estimate the firm-product match quality by taking the time average of the residuals.

These specifications include

$$y_{ipjt} = X_{ipjt}\gamma + \delta_{pt} + \delta_{it} + \varepsilon_{ipjt} \quad (\text{C.35})$$

$$y_{ipjt} = X_{ipjt}\gamma_1 + X_{ijt}\gamma_2 + X_{pt}\gamma_3 + \delta_p + \delta_i + \delta_t + \varepsilon_{ipjt} \quad (\text{C.36})$$

$$y_{ipjt} = X_{ipjt}\gamma_1 + X_{ijt}\gamma_2 + \delta_{pt} + \delta_i + \delta_t + \varepsilon_{ipjt}. \quad (\text{C.37})$$

A critical limitation of this approach is the lack of clarity regarding what the time-varying residuals capture, and whether their average offers a more accurate measure of firm-product match quality than the baseline model.